

UNIT-I

INTRODUCTION TO ENGINEERING DRAWING

Lay out of a drawing sheet:-

Every drawing sheet is to follow a particular layout. As a standard practice sufficient margins are to be provided on all sides of the drawing sheet. The drawing sheet should have drawing space and title page. A typical layout of a drawing sheet is shown in the figure below:

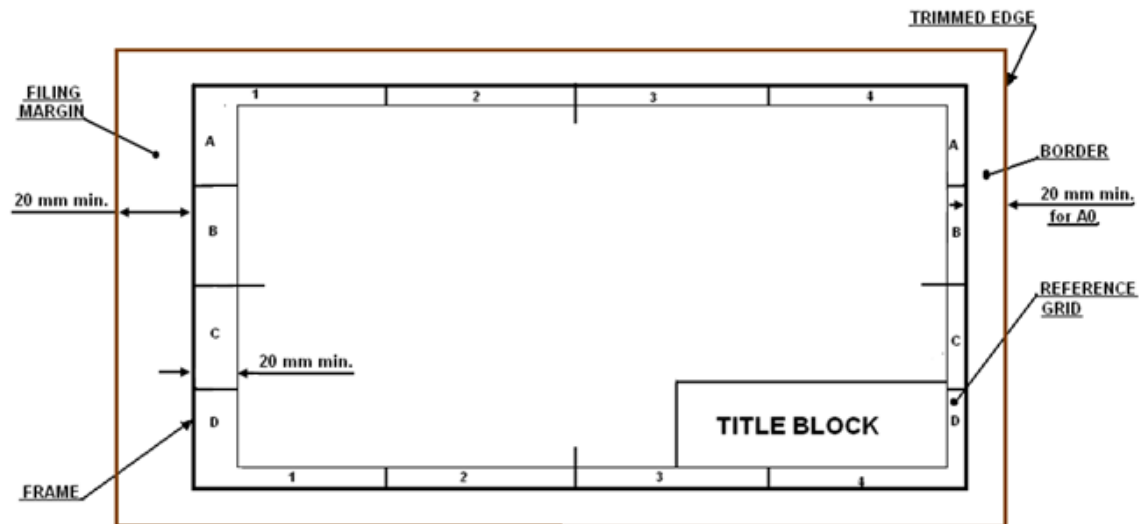


Figure 1. A typical layout of a drawing sheet.

- Borders – A minimum of 10 mm space left all around in between the trimmed edges of the sheet.
- Filing margin – Minimum 20 mm space left on the left hand side with border included. This provided for taking perforations.
- Grid reference system – This is provided on all sizes of industrial drawing sheets for easy location of drawing within the frame. The length and the width of the frames are divided into even number of divisions and labeled using numerals or capital letters. Number of divisions for a particular sheet depends on complexity of the drawing. The grids along the horizontal edges are labeled in numerals whereas grids along vertical edges are labeled using capital letters. The length of each grids can be between 25 mm and 75 mm. Numbering and lettering start from the corner of the sheet opposite to the title box and are repeated on the opposite sides. They are written upright. Repetition of letters or numbers like AA, BB, etc., if they exceed that of the alphabets. For first year engineering students grid references need not be followed.
- Title box – An important feature on every drawing sheet. This is located at the bottom right hand corner of every sheet and provides the technical and administrative details of the drawing. The title box is divided into two zones
- Identification zone: In this zone the details like the identification number or part number, Title of the drawing, legal owner of the drawing, etc. are to be mentioned.
- Additional information zone: Here indicative items like symbols indicating the system of projection, scale used, etc., the technical items like method of surface texture, tolerances, etc., and other administrative items are to be mentioned.

Layout of the title box recommended for engineering Drawing course-

The title box shown in figure 2 can be used for the engineering Drawing Course.

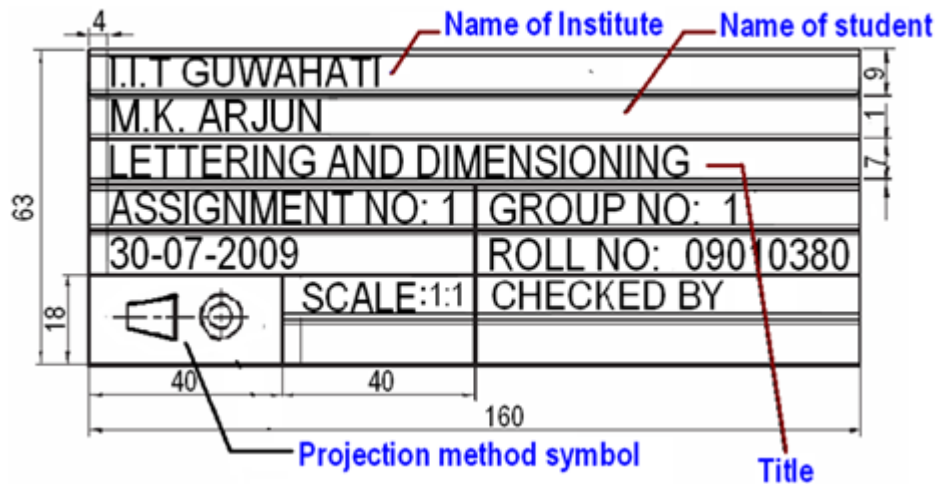


Figure 2. A typical title box recommended for engineering students.

Lettering

Lettering is used for writing of titles, sub-titles, dimensions, scales and other details on a drawing. Typical lettering features used for engineering drawing is shown in figure 3. The following rules are to be followed in lettering. The letter sizes generally recommended for various items are shown in Table 1.

- Essential features of lettering – legibility, uniformity, ease, rapidity, and suitability for microfilming/photocopying/any other photographic processes
- No ornamental and embellishing style of letter
- Plain letters and numerals which are clearly distinguishable from each other in order to avoid any confusion even in case of slight mutilations
- The Indian standard followed for lettering is BIS: 9609
- Single stroke lettering for use in engineering drawing – width of the stem of the letters and numerals will be uniformly thick equal to thickness of lines produced by the tip of the pencil.
- Single stroke does not mean – entire letter written without lifting the pencil/pen
- Lettering types generally used for creating a drawing are
- Lettering A – Height of the capital letter is divided into 14 equal parts
- Lettering B – Height of the capital letter is divided into 10 equal parts
- Table 2 and Table 3 indicates the specifications for Type A and Type B letters.

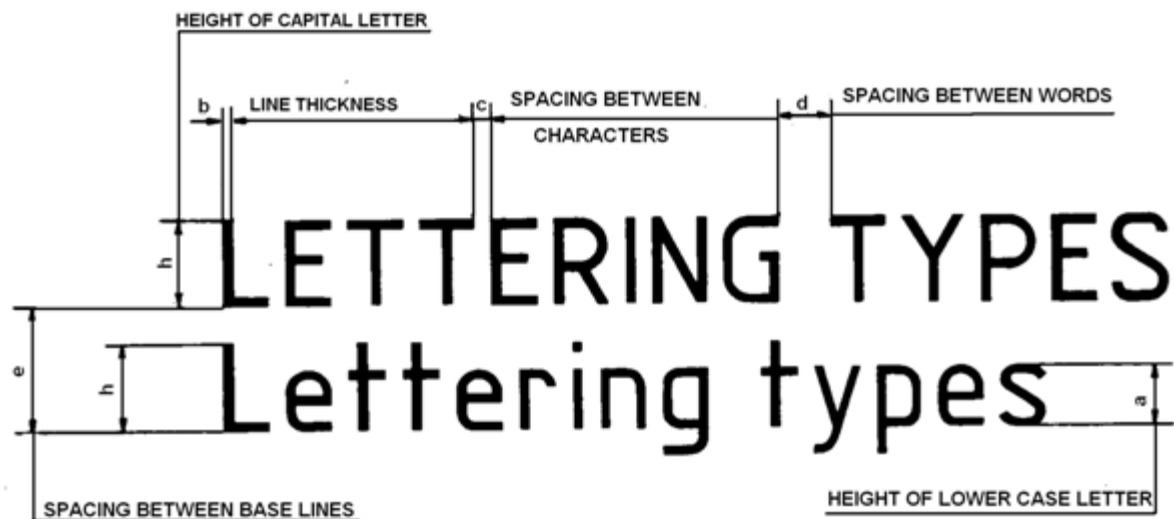


Figure 3. Typical lettering features.

Heights of Letters and Numerals

- Height of the capital letters is equal to the height of the numerals used in dimensioning
- Height of letters and numerals – different for different purposes

Table 1 the letter sizes recommended for various items

Sr. No.	Item	Size (mm)
1	Name of the company	10, 14, 20
2	Drawing numbers, letters denoting section planes	10, 14
3	Title of the Drawing	7, 10
4	Sub-titles and heading	5, 7
5	Dimensioning, Notes, Schedules, Material list	3.5, 7
6	Alteration entries and tolerances	3.5

Table 2. Specifications of A -Type Lettering

Specifications	Value	Size (mm)						
Capital letter height	h	2.5	3.5	5	7	10	14	20
Lowercase letter height	$a = (5/7)h$	-	2.5	3.5	5	7	10	14
Thickness of lines	$b = (1/14)h$	0.18	0.25	0.35	0.5	0.7	1	1.4
Spacing between characters	$c = (1/7)h$	0.35	0.5	0.7	1	1.4	2	2.8
Min. spacing b/n words	$d = (3/7)h$	1.05	1.5	2.1	3	4.2	6	8.4
Min. spacing b/n baselines	$e = (10/7)h$	3.5	5	7	10	14	20	28

Table 3. Specifications of B -Type Lettering

Specifications	Value	Size (mm)						
Capital letter height	h	2.5	3.5	5	7	10	14	20
Lowercase letter height	$a = (7/10)h$	-	2.5	3.5	5	7	10	14
Thickness of lines	$b = (1/10)h$	0.25	0.35	0.5	0.7	1	1.4	2
Spacing between characters	$c = (1/5)h$	0.5	0.7	1	1.4	2	2.8	4
Min. spacing b/n words	$d = (3/5)h$	1.5	2.1	3	4.2	6	8.4	12
Min. spacing b/n baselines	$e = (7/5)h$	3.5	5	7	10	14	20	28

How to begin drawing?

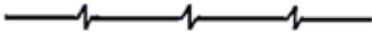
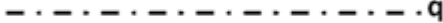
- To start with the preparation of a drawing the procedure mentioned below may be followed:
- Clean the drawing board and all the drawing instruments using duster.
- Fix the drawing sheet on the drawing board.
- Fix the mini-drafter in a convenient position.
- Draw border lines using HB pencil.
- Complete the title box using HB pencil.

- Plan spacing of drawings b/n two problems/views beforehand.
- Print the problem number on the left top and then commence the drawing work
- Keeping the drawing clean is a must
- Never sharpen pencils over drawing.
- Clean pencil point with a soft cloth after sharpening.
- Keep drawing instruments clean.
- Rest hands on drawing instruments as much as possible – to avoid smearing the graphite on the drawing.
- When darkening lines – try to work from the top of the drawing to the bottom, and from left to the right across the drawing.
- Use brush to remove eraser particles. Never use hands.
- Always use appropriate drawing pencils.

Lines

Lines is one important aspect of technical drawing. Lines are always used to construct meaningful drawings. Various types of lines are used to construct drawing, each line used in some specific sense. Lines are drawn following standard conventions mentioned in BIS (SP46:2003). A line may be curved, straight, continuous, segmented. It may be drawn as thin or thick. A few basic types of lines widely used in drawings are shown in Table 1.

Table 1. Types of letters used in engineering drawing.

Illustration	Application
Thick 	Outlines, visible edges, surface boundaries of objects, margin lines
Continuous thin 	Dimension lines, extension lines, section lines leader or pointer lines, construction lines, boarder lines
Continuous thin wavy 	Short break lines or irregular boundary lines – drawn freehand
Continuous thin with zig-zag 	Long break lines
Short dashes, gap 1, length 3 mm 	Invisible or interior surfaces
Short dashes 	Center lines, locus lines Alternate long and short dashes in a proportion of 6:1,
Long chain thick at end and thin elsewhere 	Cutting plane lines

Line Strokes- Line strokes refer to the directions of drawing straight and curved lines. The standards for lines IS given in BIS: SP-46, 2003 Vertical and inclined lines are drawn from top to bottom, horizontal lines are drawn from left to right. Curved lines are drawn from left to right or top to bottom. The direction of strokes are illustrated in figure 1.

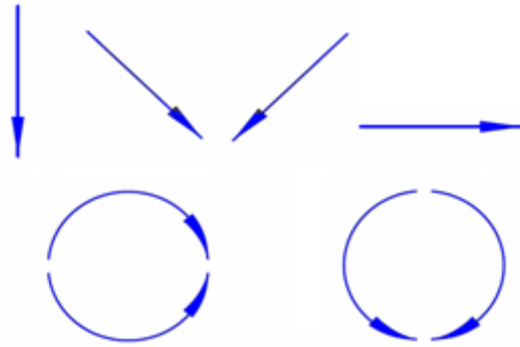


Figure 1. The line strokes for drawing straight and curved lines.

Conventions used in lines

- International systems of units (SI) – which is based on the meter.
- Millimeter (mm) - The common SI unit of measure on engineering drawing.
- Individual identification of linear units is not required if all dimensions on a drawing are in the same unit (mm).
- The drawing should contain a note: ALL DIMENSIONS ARE IN MM. (Bottom left corner outside the title box)

Typical figures showing various lines used in the construction of engineering drawing is shown in figure 2.

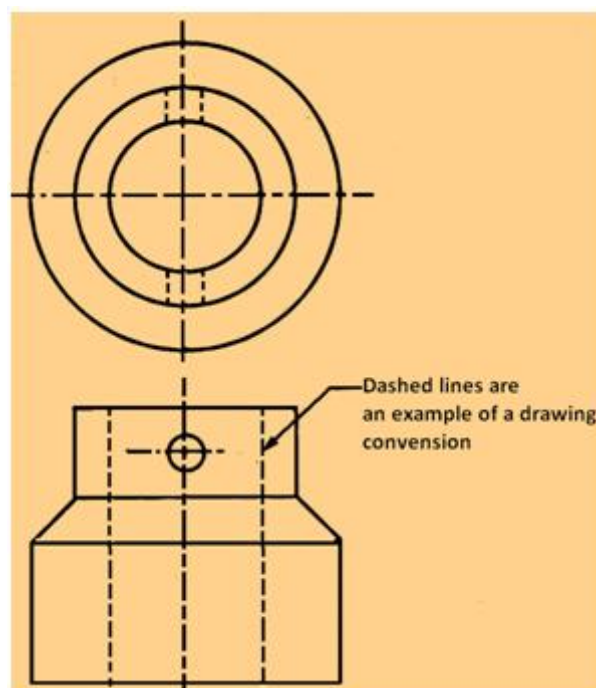
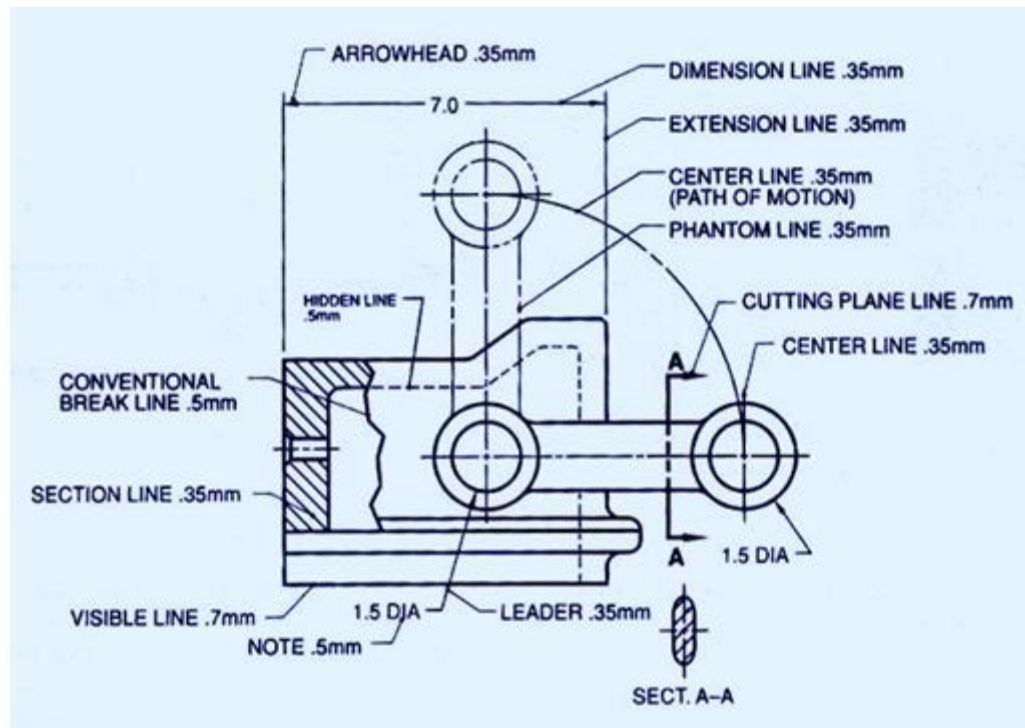


Figure 2 Typical figure showing various lines used engineering drawing

A typical use of various lines in an engineering drawing is shown in figure below:



Dimensioning:

The size and other details of the object essential for its construction and function, using lines, numerals, symbols, notes, etc are required to be indicated in a drawing by proper dimensioning. These dimensions indicated should be those that are essential for the production, inspection and functioning of the object and should be mistaken as those that are required to make the drawing of an object. The dimensions are written either above the dimension lines or inserted at the middle by breaking the dimension lines.

Normally two types of dimensioning system exist. i.e., aligned system and the unidirectional system. These are shown in figure 3.

In the aligned system the dimensions are placed perpendicular to the dimension line in such a way that it may be read from bottom edge or right hand edge of the drawing sheet. The horizontal and inclined dimension can be read from the bottom whereas all the vertical dimensions can be read from the right hand side of the drawing sheet.

In the unidirectional system, the dimensions are so oriented such that they can be read from the bottom of the drawing.

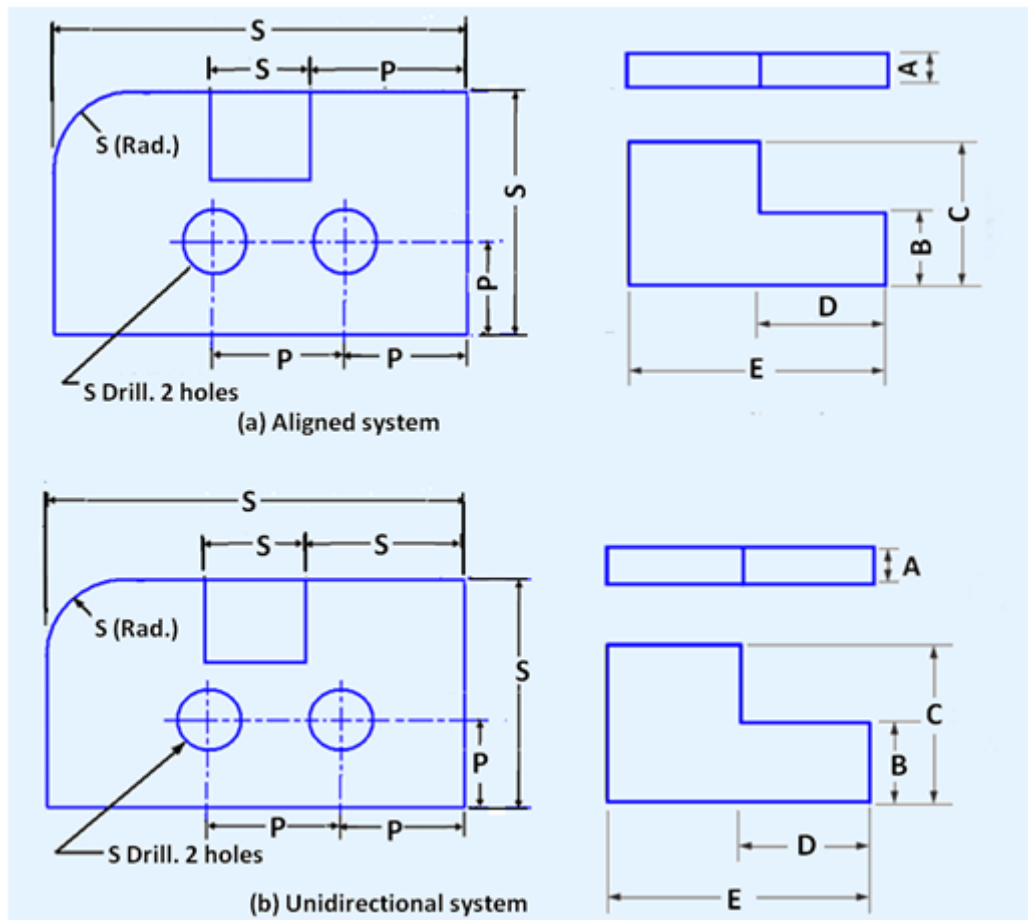


Figure 3. The aligned system and unidirectional system of dimensioning.

Rules to be followed for dimensioning. Refer figure 4.

- Each feature is dimensioned and positioned only once.
- Each feature is dimensioned and positioned where its shape shows.
- Size dimensions – give the size of the component.
- Every solid has three dimensions, each of the geometric shapes making up the object must have its height, width, and depth indicated in the dimensioning.

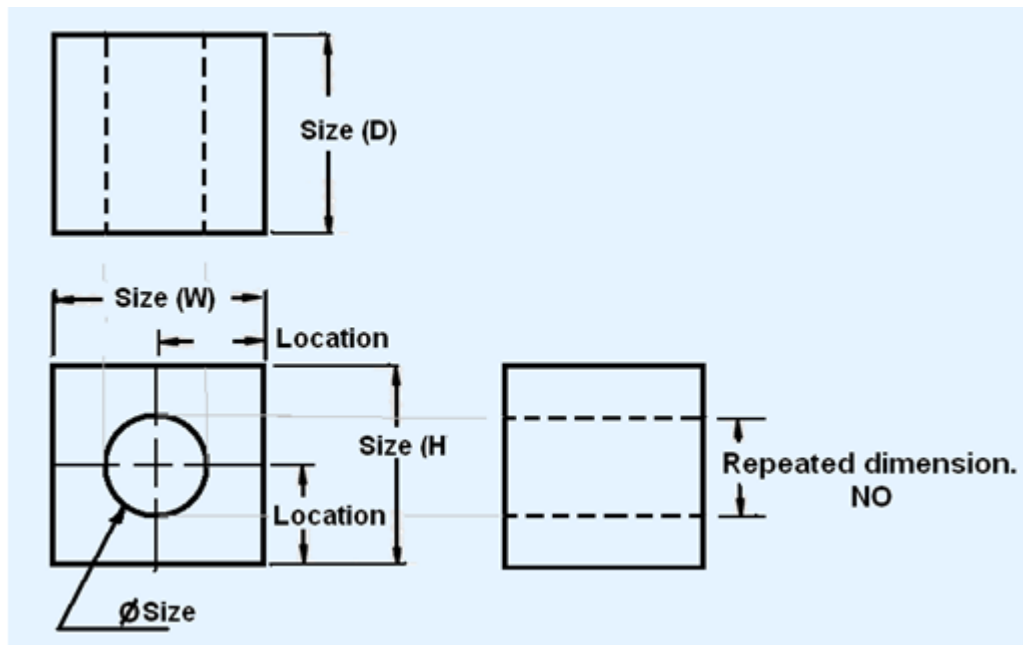


Figure 4. Typical dimension lines

Dimensioning consists of the following:

- A thin, solid line that shows the extent and direction of a dimension. Dimension lines are broken for insertion of the dimension numbers
- Should be placed at least 10 mm away from the outline and all
- other parallel dimensions should be at least 6 mm apart, or more, if space permits

The important elements of dimensioning consists of extension lines, leader line, arrows and dimensions.

Extension line – a thin, solid line perpendicular to a dimension line, indicating which feature is associated with the dimension. There should be a visible gap of 1.5 mm between the feature's corners and the end of The extension Line figure 5 shows extension Lines

Leader Line

A thin, solid line used to indicate the feature with which a dimension, note, or symbol is associated. Generally this is a straight line drawn at an angle that is neither horizontal nor vertical. Leader line is terminated with an arrow touching the part or detail. On the end opposite the arrow, the leader line will have a short, horizontal shoulder. Text is extended from this shoulder such that the text height is centered with the shoulder line

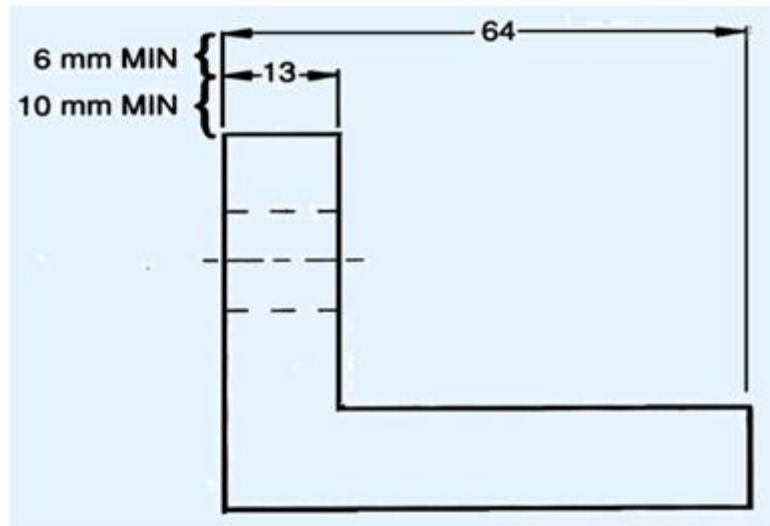
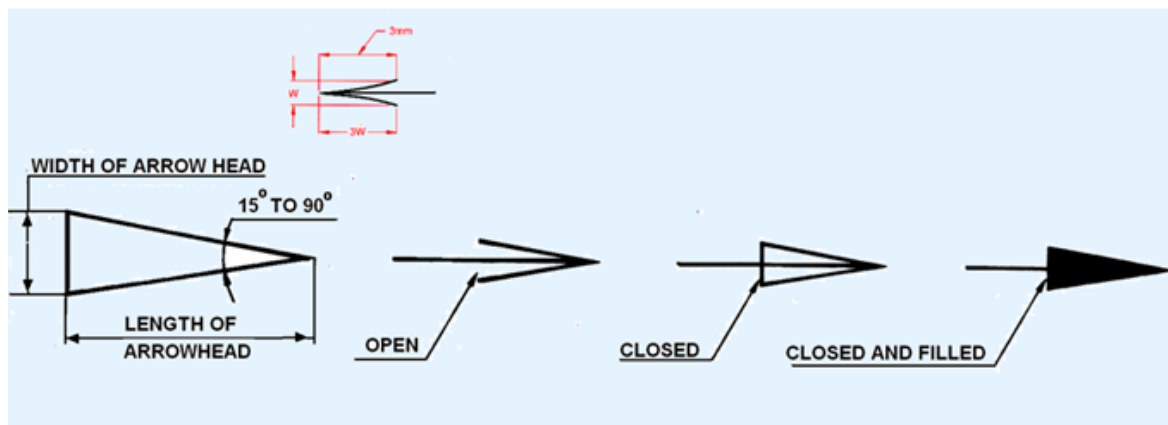


Figure 5. Showing extension lines

- Arrows – 3 mm wide and should be 1/3rd as wide as they are long - symbols placed at the end of dimension lines to show the limits of the dimension. Arrows are uniform in size and style, regardless of the size of the drawing. Various types of arrows used for dimensioning is shown in figure 6.



The specification of dimension lines are shown in figure 7.

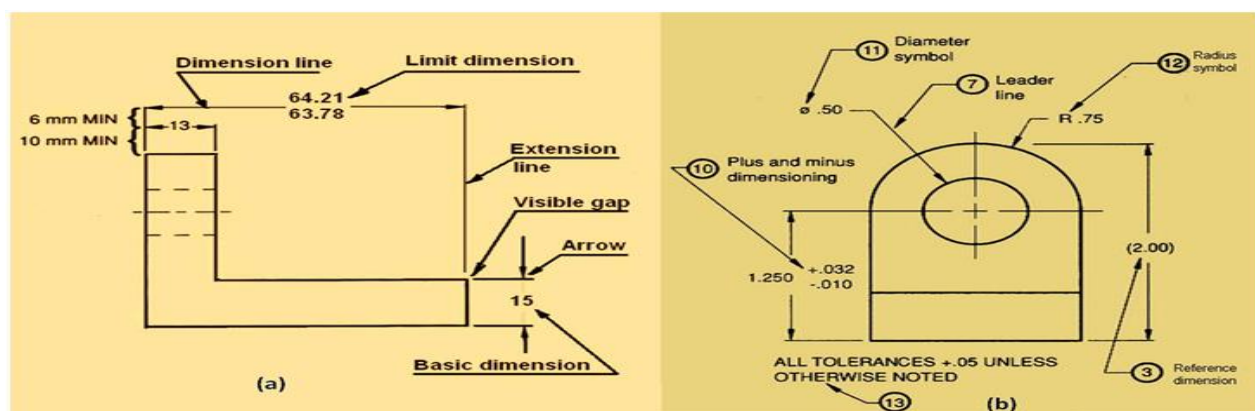


Figure 7 showing the specification of dimension lines.

Dimensioning of angles: The normal convention for dimensioning of angles are illustrated in figure 8.

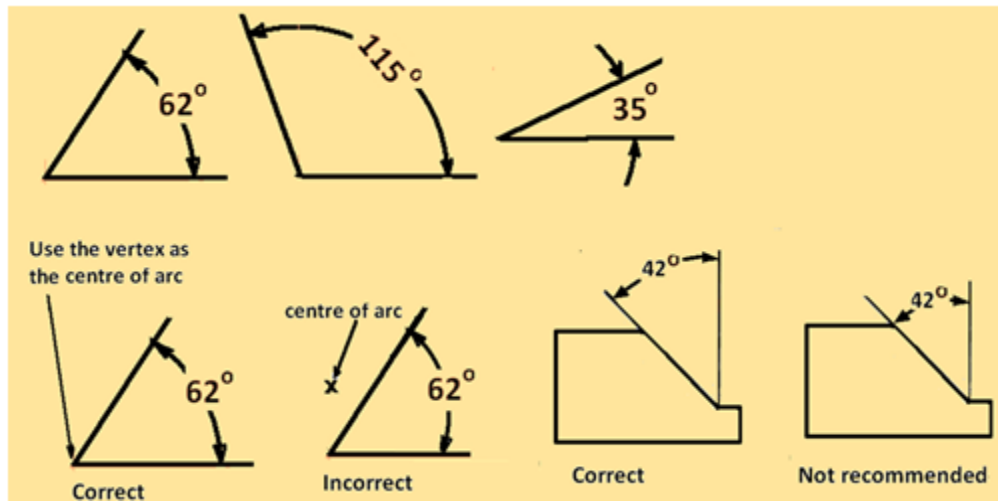


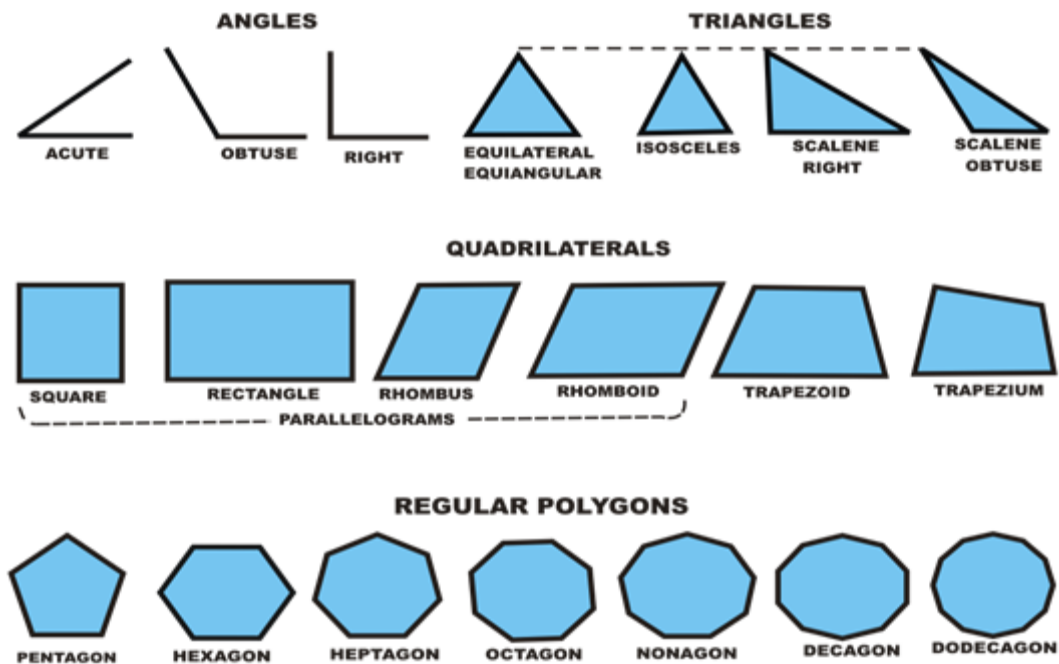
Figure 8 conventions used for dimensioning angles.

RULES OF DIMENSIONING

- Between any two extension lines, there must be one and only one dimension line bearing one dimension.
- As far as possible, all the dimensions should be placed outside the views. Inside dimensions are preferred only if they are clearer and more easily readable.
- All the dimensions on a drawing must be shown using either Aligned System or Unidirectional System. In no case should, the two systems be mixed on the same drawing.
- The same unit of length should be used for all the dimensions on a drawing. The unit should not be written after each dimension, but a note mentioning the unit should be placed below the drawing.
- Dimension lines should not cross each other. Dimension lines should also not cross any other lines of the object.
- All dimensions must be given.
- Each dimension should be given only once. No dimension should be redundant.
- Do not use an outline or a centre line as a dimension line. A centre line may be extended to serve as an extension line.
- Void dimensioning hidden lines.
- For dimensions in series, adopt any one of the following ways.
 - Chain dimensioning (Continuous dimensioning) all the dimensions are aligned in such a way that an arrowhead of one dimension touches tip-to-tip the arrowhead of the adjacent dimension. The overall dimension is placed outside the other smaller dimensions.
 - Parallel dimensioning (Progressive dimensioning) All the dimensions are shown from a common reference line. Obviously, all these dimensions share a common extension line. This method is adopted when dimensions have to be established from a particular datum surface
 - Combined dimensioning. When both the methods, i.e., chain dimensioning and parallel dimensioning are used on the same drawing, the method of dimensioning is called combined dimensioning.

Geometric Construction

- Drawing consists of construction of primitive geometric forms viz. Points, lines and planes that serve as building blocks for more complicated geometric shapes and defining the position of object in space.
- The use of lines for obtaining the drawing of planes is shown in figure 1.



- Figure 1 illustrates various planes generally encountered
- Solids are obtained by combination of planes. Plane surfaces of simple solids are shown in figure 2.

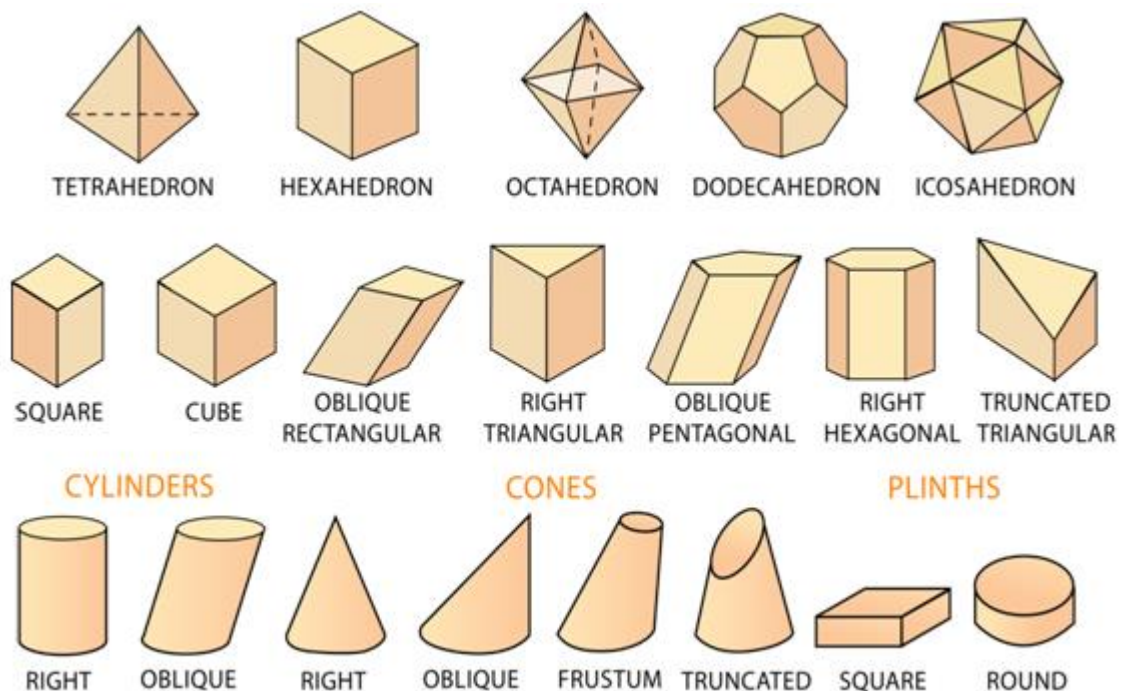


Figure 2 surfaces of few simple solids.

In addition curved surfaces also exist. Figure 3 shows some of the solids having curved surfaces.

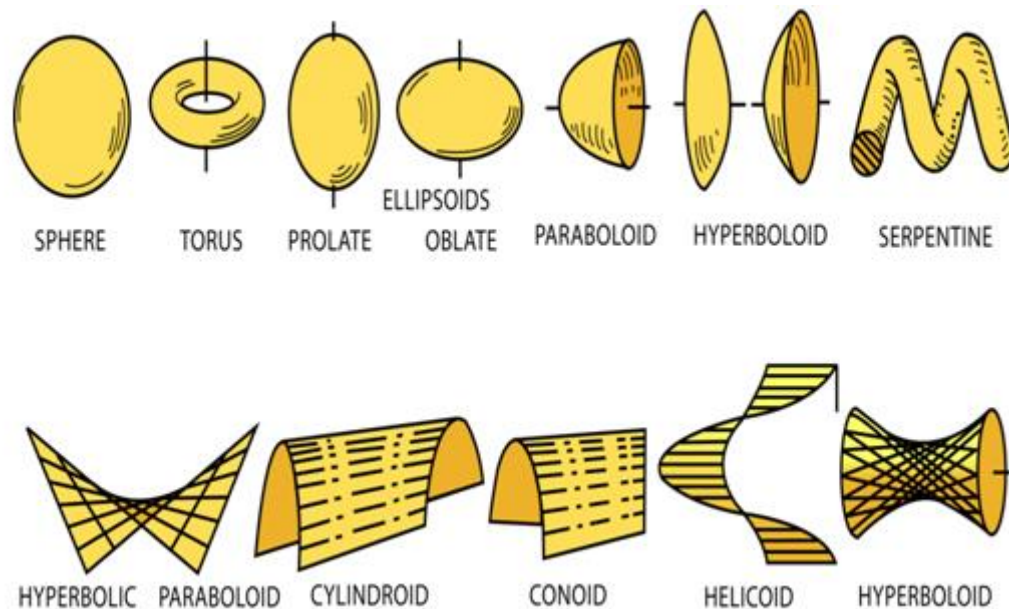


Figure 3. Solids having curved surfaces.

Primitive geometric forms

The shapes of objects are formed from primitive geometric forms. These are

- Point
- Line
- Plane
- Solid
- Doubly curved surface and object

Warped surface

The basic 2-D geometric primitives, from which other more complex geometric forms are derived.

- Points,
- Lines,
- Circles, and
- Arcs

Point

A point is a theoretical location that has neither width, height, nor depth and describes exact location in space. A point is represented in technical drawing as a small cross made of dashes that are approximately *3 mm long*. As shown in figure 4, a point is used to mark the locations of centers and loci, the intersection ends, middle of entities

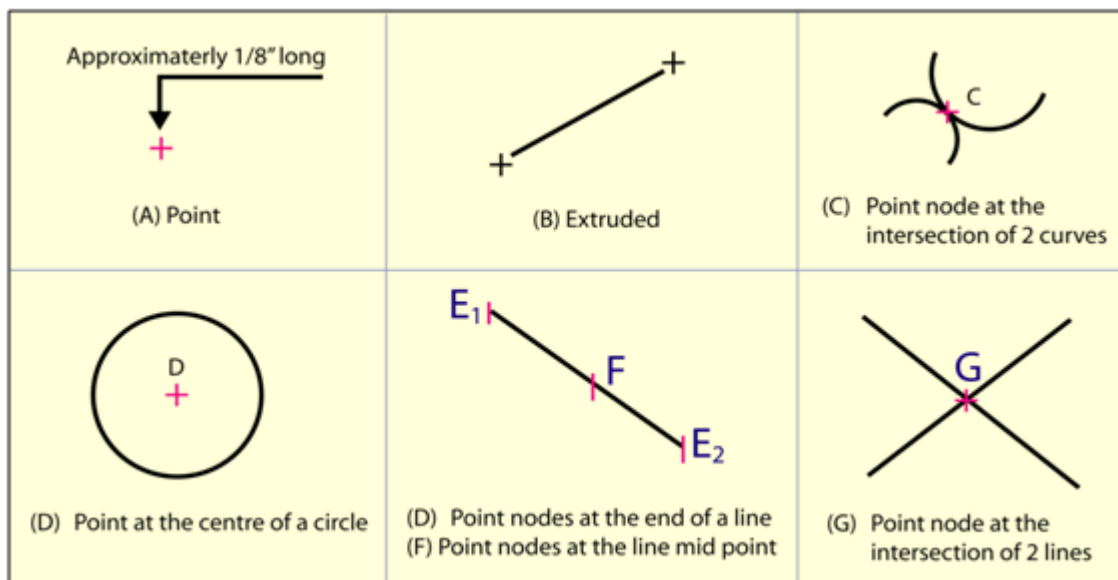


Figure 4. Shows the various use of points.

Line

A line is a geometric primitive that has length and direction, but no thickness. Lines may be straight, curved or a combination of these. As shown in figure 5, lines have few important relationship or conditions, such as parallel, intersecting, and tangent. Lines can be of specific length or non-specific length. A Ray is a straight line that extends to infinity from a specified point.

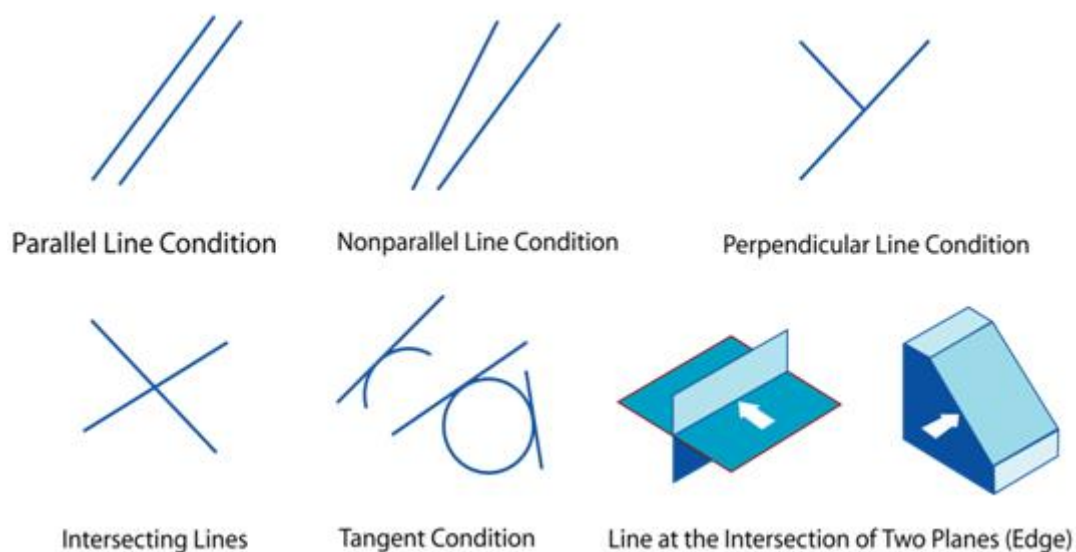


Figure 5. Relationship of one line to another line or arc

Bisecting a line

The procedure of bisecting a given line AB is illustrated in figure 6. With A as center and radius equal to higher than half AB, draw two arcs. With B as center and with the same radius draw another arc intersecting the previous arcs. The line joining the intersection points is the perpendicular bisector of the line AB.

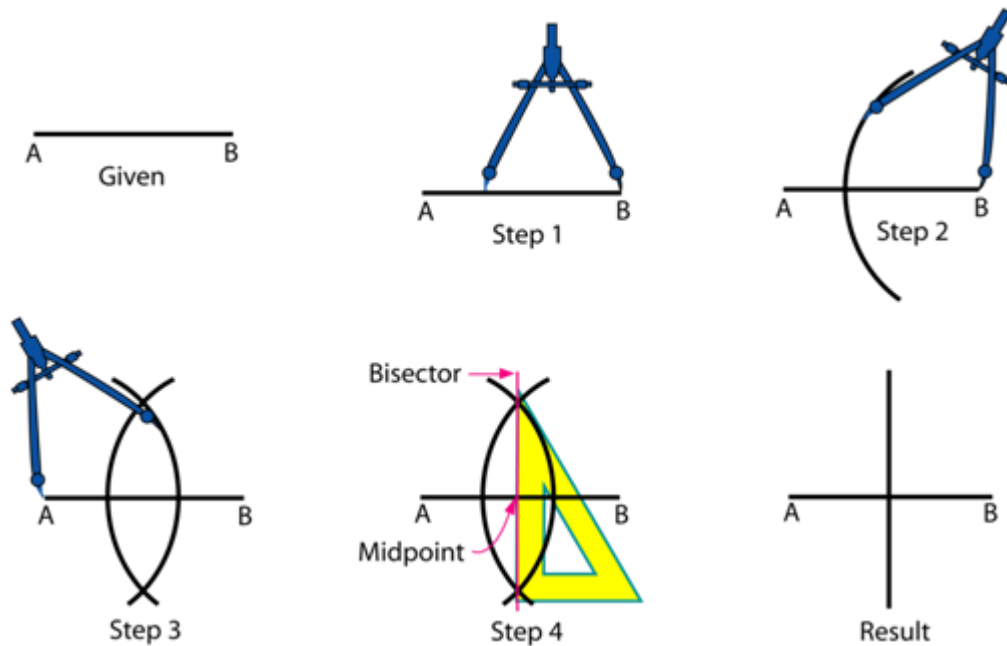


Figure 6. Illustrates the method of bisecting a line

Dividing a line into equal parts

The method of dividing a line MO into equal number of parts is illustrated in figure 7.

- Draw a line MO at any convenient angle (preferably an acute angle) from point M.
- From M and along MO, cut off with a divider equal divisions (say three) of any convenient length.
- Draw a line joining ON.
- Draw lines parallel to MO through the remaining points on line MO.

The intersection of these lines with line MN will divide the line into (three) equal parts.

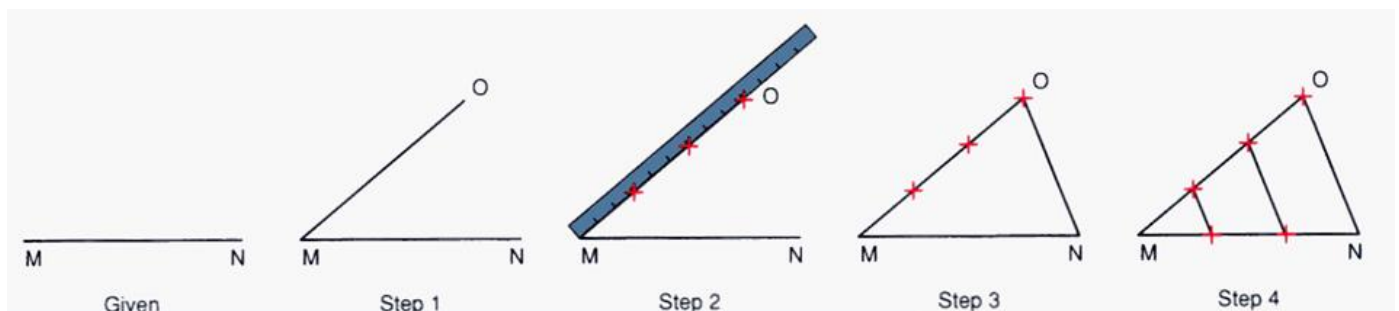


Figure 7. Dividing a line into equal number of parts

Planar tangent condition exists when two geometric forms meet at a single point and do not intersect. This is self-explanatory from figure 8.

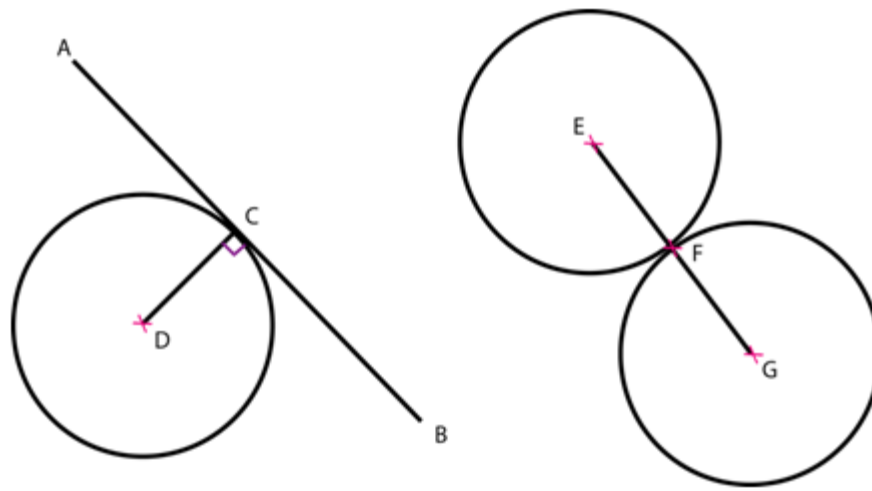


Figure 8. Illustrates the existence of planar tangent condition.

Locating tangent points on circle and arcs

The method of locating tangent points on circle and arcs as well as the common tangent to two circles are shown in figure 9(a) and (b) .

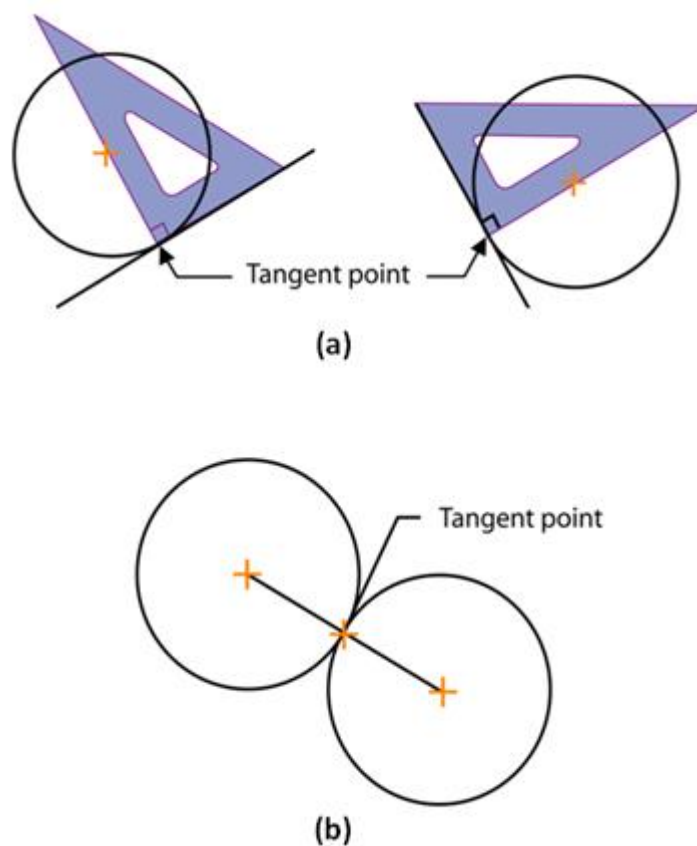


Figure 9. Locating the tangent points to arcs or circles.

Drawing an arc tangent to a given point on the line

The steps for drawing the arc tangent to a given point on a line is shown in figure 10.

Given line AB and tangent point T. Construct a line perpendicular to line AB and through point T.

Locate the center of the arc by making the radius on the perpendicular line. Put the point of the compass at the center of the arc, set the compass for the radius of the arc, and draw the arc which will be tangent to the line through the point T.

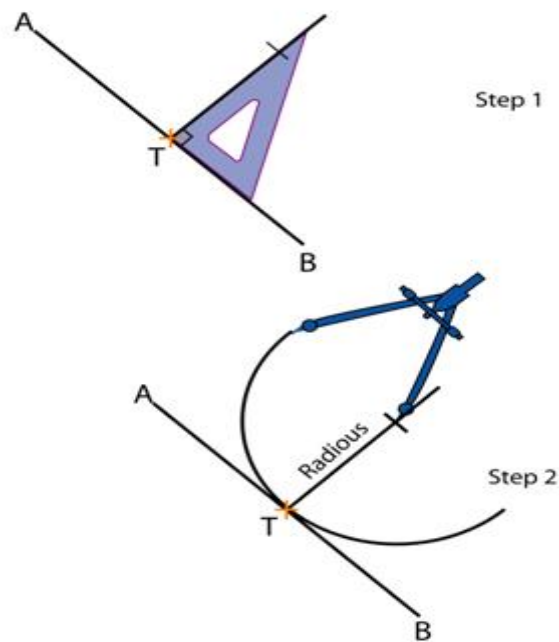


Figure 10. Drawing an arc tangent to the a given point on a line

Drawing an arc, tangent to two lines

The steps used to drawn an arc tangent to two lines is illustrated in figure 11.

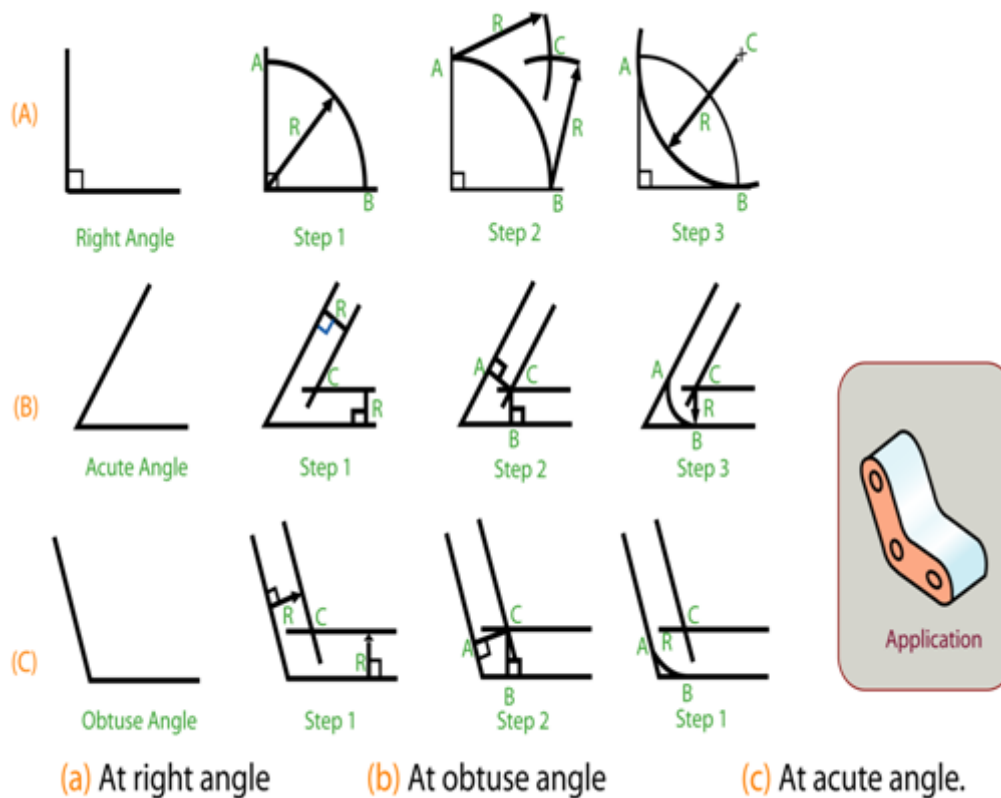


Figure 11. Illustrates the method of drawing an arc tangent to two lines.

Drawing an arc, tangent to a line and an arc

Figure 12 shows the steps in drawing an arc tangent to a line and an arc that (a) that do not intersect and (b) that intersect each other.

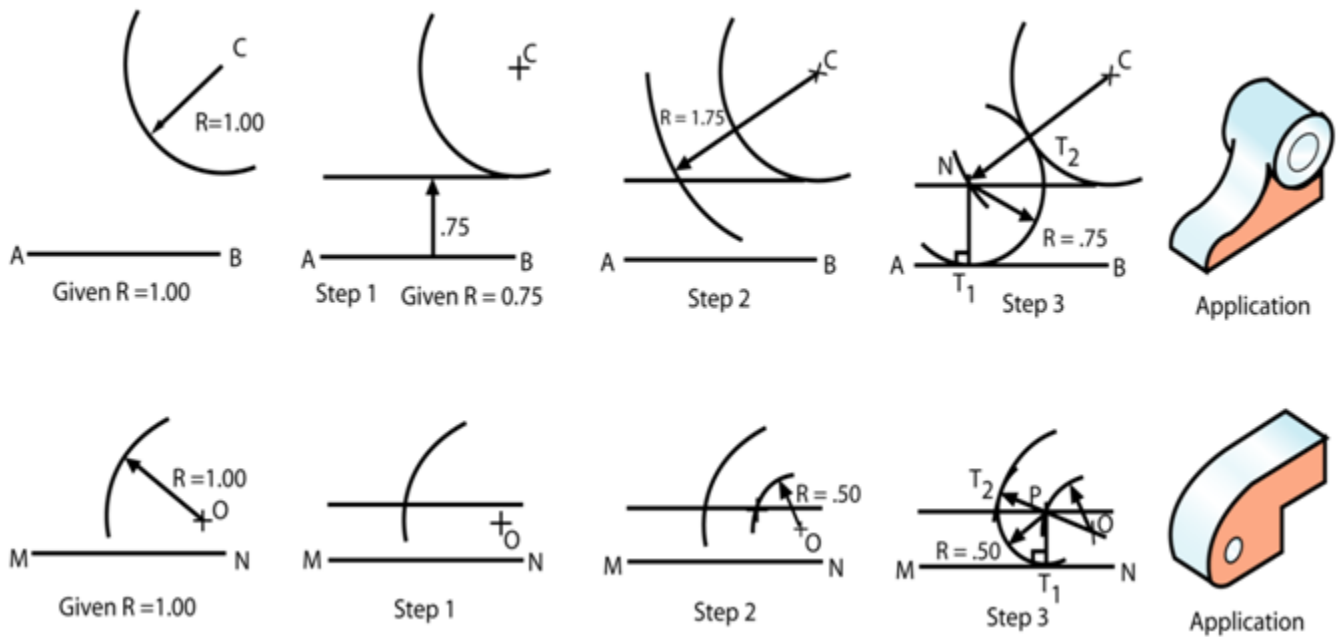


Figure 12 drawing an arc tangent to a line and an arc

Construction of Regular Polygon of given length AB

To construct a regular polygon with length of edge AB us shown in figure 1.

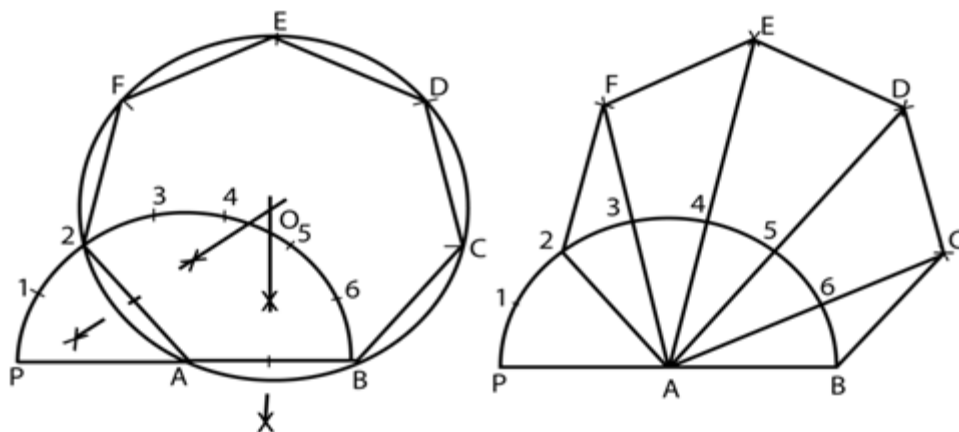


Figure 1. Construction of a regular polygon with a given length of edge.

- Draw a line of length AB. With A as centre and radius AB, draw a semicircle.
- With the divider, divide the semicircle into the number of sides (example of number of side 7 is shown in figure 1) of the polygon.
- Draw a line joining A with the second division-point 2.
- The perpendicular bisectors of A2 and AB meet at O. Draw a circle with centre O and radius OA.

- With length A2, mark points F, E, D & C on the circumferences starting from 2 (Inscribe circle method)
- With centre B and radius AB draw an arc cutting the line A6 produced at C. Repeat this for other points D, E & F (Arc method)

General method of drawing any polygon

- A more general method of drawing any polygon with a given length of edge is shown in figure 2.
- Draw AB = given length of polygon
- At B, Draw BP perpendicular & = AB
- Draw Straight line AP
- With center B and radius AB, draw arc AP.
- The perpendicular bisector of AB meets the line AP and arc AP in 4 and 6 respectively.
- Draw circles with centers as 4, 5, & 6 and radii as 4B, 5B, & 6B and inscribe a square, pentagon, & hexagon in the respective circles.
- Mark point 7, 8, etc with 6-7, 7-8, etc. = 4-5 to get the centers of circles of heptagon and octagon, etc.

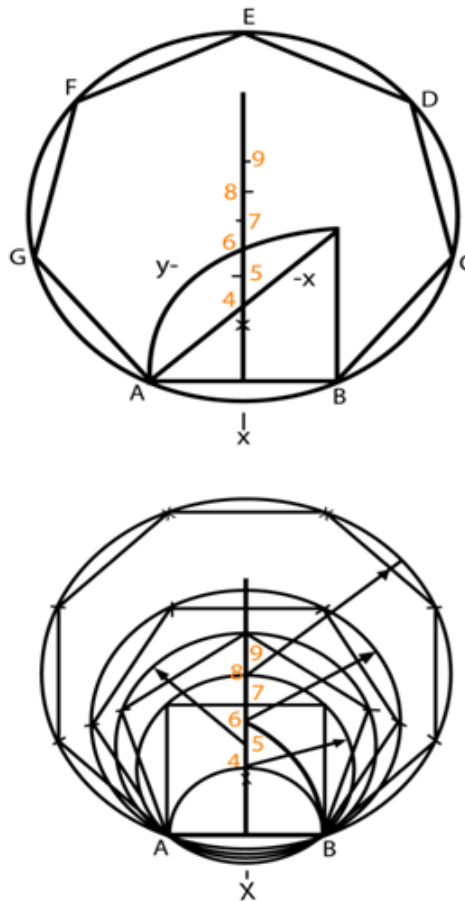


Figure 2 drawing any polygon with a given length of edge

Inscribe a circle inside a regular polygon

The method of inscribing a circle inside a regular polygon is illustrated in figure 3.

- Bisect any two adjacent internal angles of the polygon.

- From the intersection of these lines, draw a perpendicular to any one side of the polygon (say OP).
- With OP as radius, draw the circle with O as center

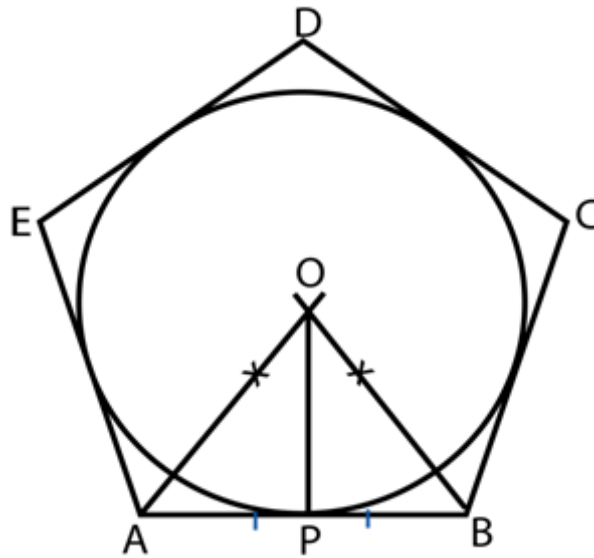


Figure 3 inscribing a circle inside a regular polygon

Inscribe a regular polygon of any number of sides (say $n = 5$), in a circle

Figure 4 shows the method of inscribing a regular polygon of any number of sides.

- Draw the circle with diameter AB.
- Divide AB in to “n” equal parts
- Number them.
- With center A & B and radius AB, draw arcs to intersect at P.
- Draw line P2 and produce it to meet the circle at C.
- AC is the length of the side of the polygon.

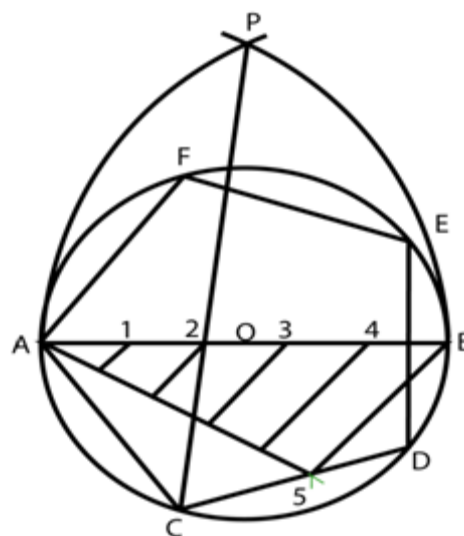


Figure 4 inscribing a regular polygon of any number of sides.

To draw a circle to touch a given line, and a given circle at a given point on it.

The method is illustrated in figure 5.

- Given: Line AB, circle with centre C and point P on the circle.
- From P, draw a tangent to the circle intersecting AB at D.
- Draw bisector of angle PDB to intersect the line through C and P at O.
- With center O and radius OP, draw the required circle.

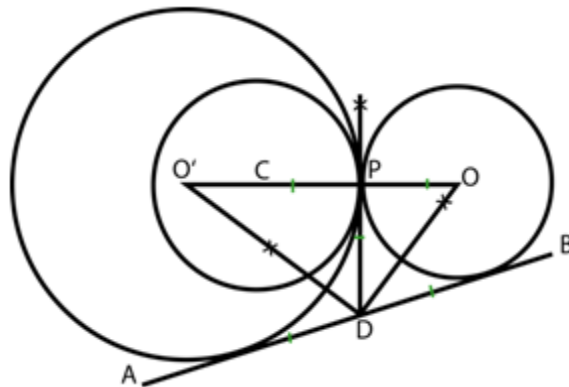


Figure 5. Shows the method of drawing a circle to touch a given line and a given circle at a particular point.

Inside a regular polygon, draw the same number of equal circles as the side of the polygon, each circle touching one side of the polygon and two of the other circles

The technique is shown in figure 6

- Draw bisectors of all the angles of the polygon, meeting at O, thus dividing the polygon into the same number of triangles.
- In each triangle inscribe a circle.

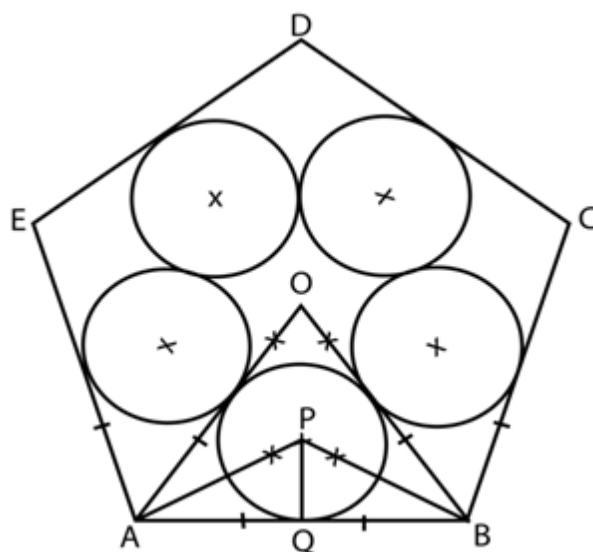


Figure 6. Drawing the same number of equal circles, in a given polygon, as the side of the polygon

Figure 7 shows the technique for drawing the same number of equal circles as the side of the polygon inside a regular polygon, each circle touching two adjacent sides of the polygon and two of the other circles.

- Draw the perpendicular bisectors of the sides of the polygon to obtain same number of quadrilaterals as the number of sides of the polygon.
- Inscribe a circle inside each quadrilateral.

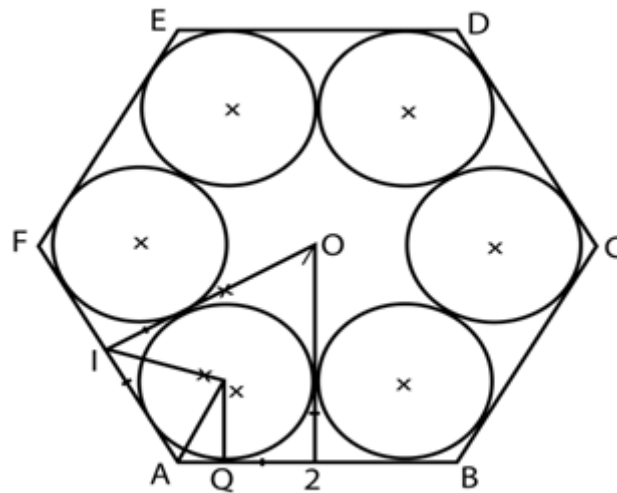


Figure 7. Drawing the same number of equal circles as the side of the polygon inside a regular polygon

Figure 8 shows the method of drawing a circle touching three lines inclined to each other but not forming a triangle.

- Let AB, BC, and AD be the lines.
- Draw bisectors of the two angles, intersecting at O.
- From O draw a perpendicular to any one line intersecting it at P.
- With O as center and OP as radius draw the desired circle.

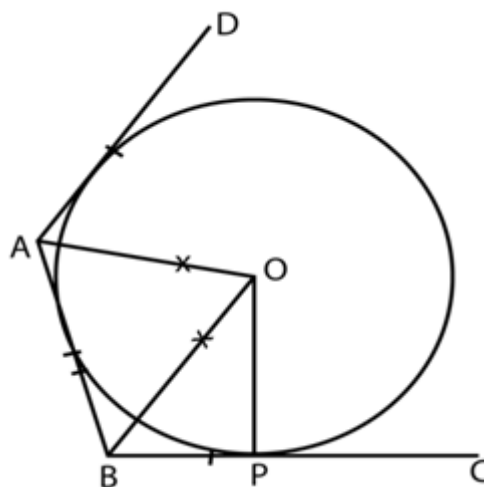


Figure 8: drawing a circle touching three lines inclined to each other

Figure 9 shows the method of drawing outside a regular polygon, the same number of equal circles as the sides of the polygon, each circle touching one side of the polygon and two of the other circles.

- Draw bisectors of two adjacent angles and produce them outside the polygon.
- Draw a circle touching the extended bisectors and the side AB (in this case) and repeat the same for other sides.

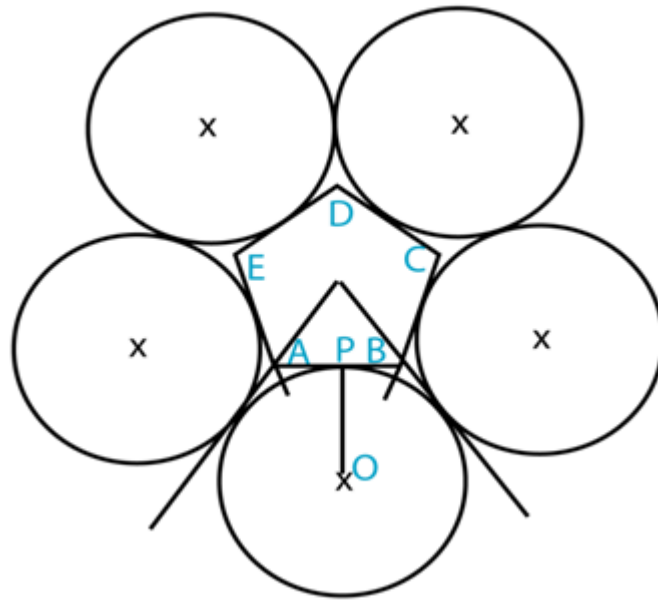


Figure 9 method of drawing outside a regular polygon, the same number of equal circles as the sides of the polygon,

Conic curves (conics)

Curves formed by the intersection of a plane with a right circular cone. E.g. Parabola, hyperbola and ellipse. Right circular cone is a cone that has a circular base and the axis is inclined at 90° to the base and passes through the center of the base. Conic sections are always "smooth". More precisely, they never contain any inflection points. This is important for many applications, such as aerodynamics, civil engineering, mechanical engineering, etc. Figure 1. Shows a right cone and the various conic curves that can be obtained from a cone by sectioning the cone at various conditions

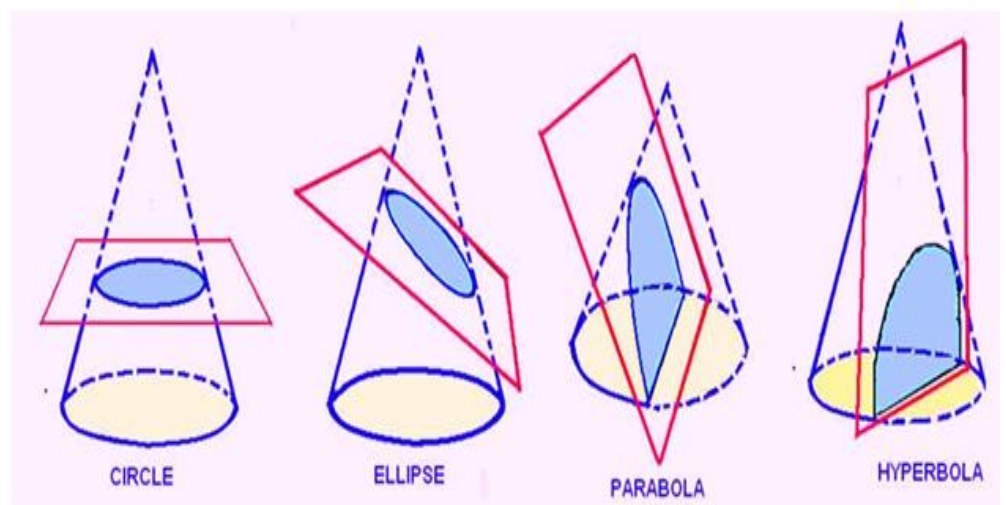
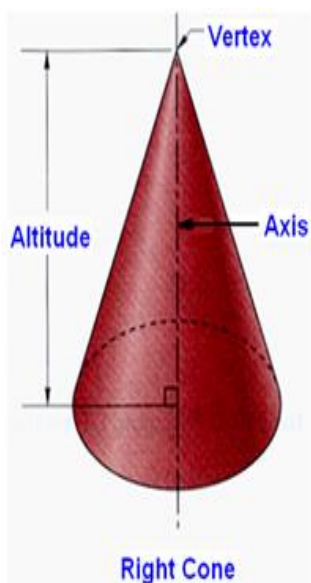


Figure 1. Shows a right cone and the various conic curves that can be obtained from a cone by sectioning the cone at various conditions.

Conic

Conic is defined as the locus of a point moving in a plane such that the ratio of its distance from a fixed point and a fixed straight line is always constant.

- Fixed point is called Focus
- Fixed line is called Directrix

This is illustrated in figure 2.

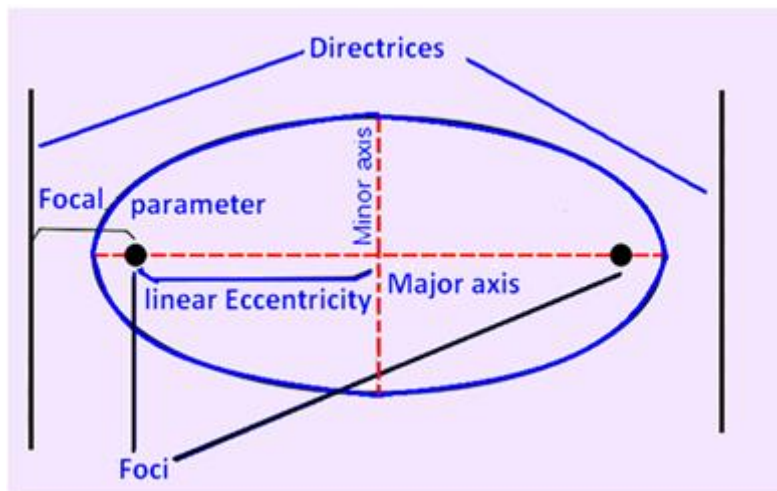


Figure 2. Illustrates the directrices and foci of a conic curve.

$$\text{Eccentricity} = \frac{\text{Distance of the point from the focus}}{\text{Distance of the point from the directrix}}$$

When

$e < 1 \rightarrow$ Ellipse

$e = 1 \rightarrow$ Parabola

$e > 1 \rightarrow$ Hyperbola

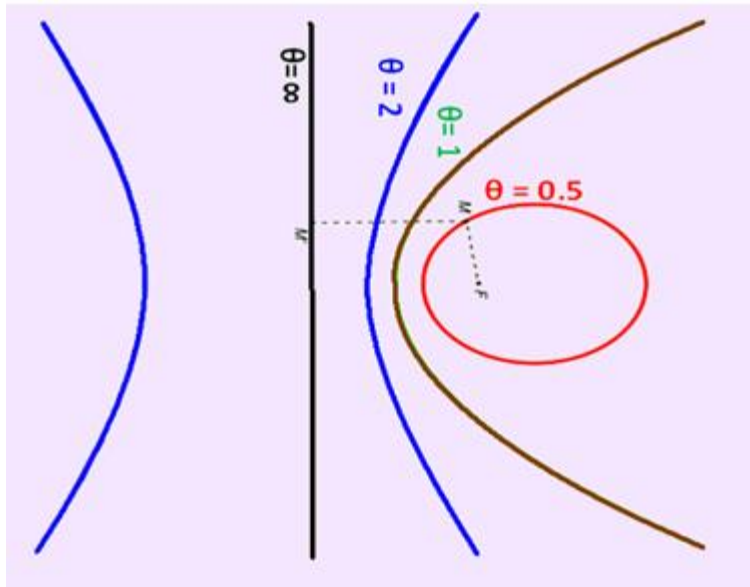


Figure 3 shows the relationship of eccentricity with different conic curves.

eg. When $e=1/2$ the curve is an Ellipse, when $e=1$, it is a parabola and when $e=2$, it is a hyperbola. Figure 3 shows the ellipse, parabola and hyperbola.

Hyperbola. Figure 3 shows the ellipse, parabola and hyperbola.

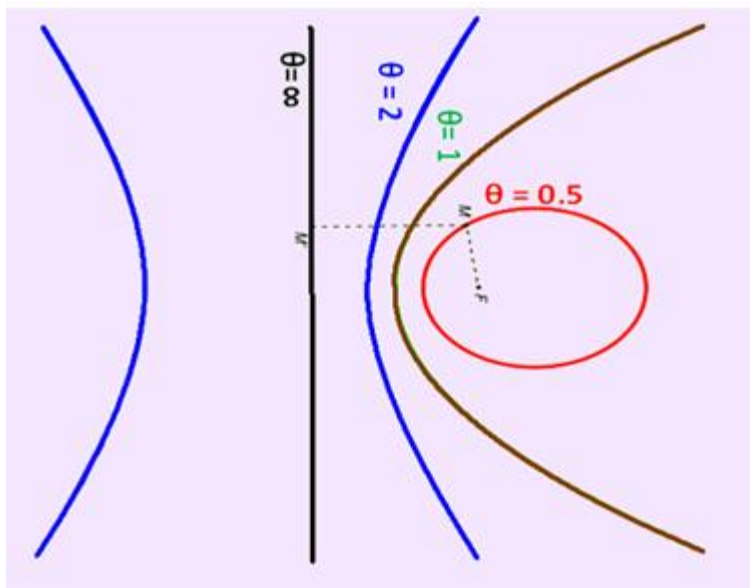


Figure 3 shows the relationship of eccentricity with different conic curves.

Ellipse

Referring to figure 4, an ellipse can be defined in the following ways.

An ellipse is obtained when a section plane, inclined to the axis of the cone, cuts all the generators of the cone.

An ellipse is the set of all points in a plane for which the sum of the distances from the two fixed points (the foci) in the plane is constant

An ellipse is also defined as a curve traced by a point, moving in a plane such that the sum of its distances from two fixed points is always the same.

Construction of Ellipse

When the distance of the directrix from the focus and eccentricity is given.

Major axis and minor axis is given.

- Arc of circle method
- Concentric circle method
- Oblong method
- Loop of the thread method

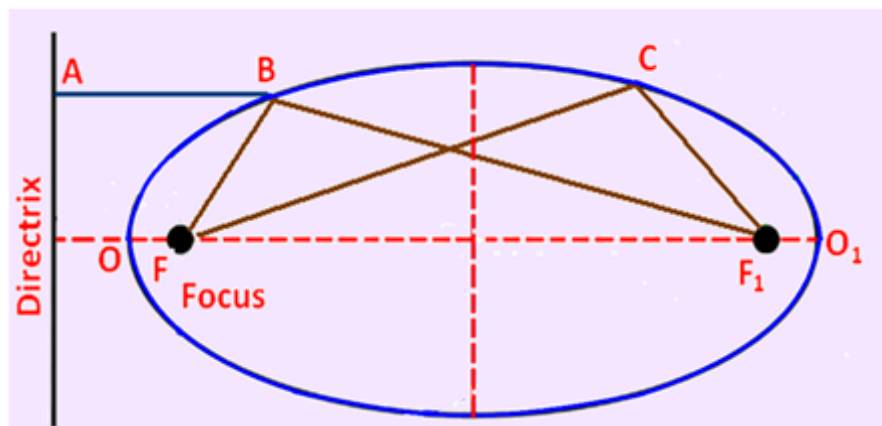


Figure 4. illustrating an ellipse.

Focus- Directrix or Eccentricity Method

Given: the distance of focus from the directrix and eccentricity

Figure 5. Shows the method of drawing an ellipse if the distance of focus from the directrix is 80 mm and the eccentricity is $\frac{3}{4}$. The distance of focus from the directrix is 80 mm and the eccentricity is $\frac{3}{4}$.

1. Draw the directrix AB and axis CC'
2. Mark F on CC' such that CF = 80 mm.
3. Divide CF into 7 equal parts and mark V at the fourth division from C. Now, $e = \frac{FV}{CV} = \frac{3}{4}$.
4. At V, erect a perpendicular VB = VF. Join CB. Through F, draw a line at 45° to meet CB produced at D. Through D, drop a perpendicular DV' on CC'. Mark O at the midpoint of V– V'.

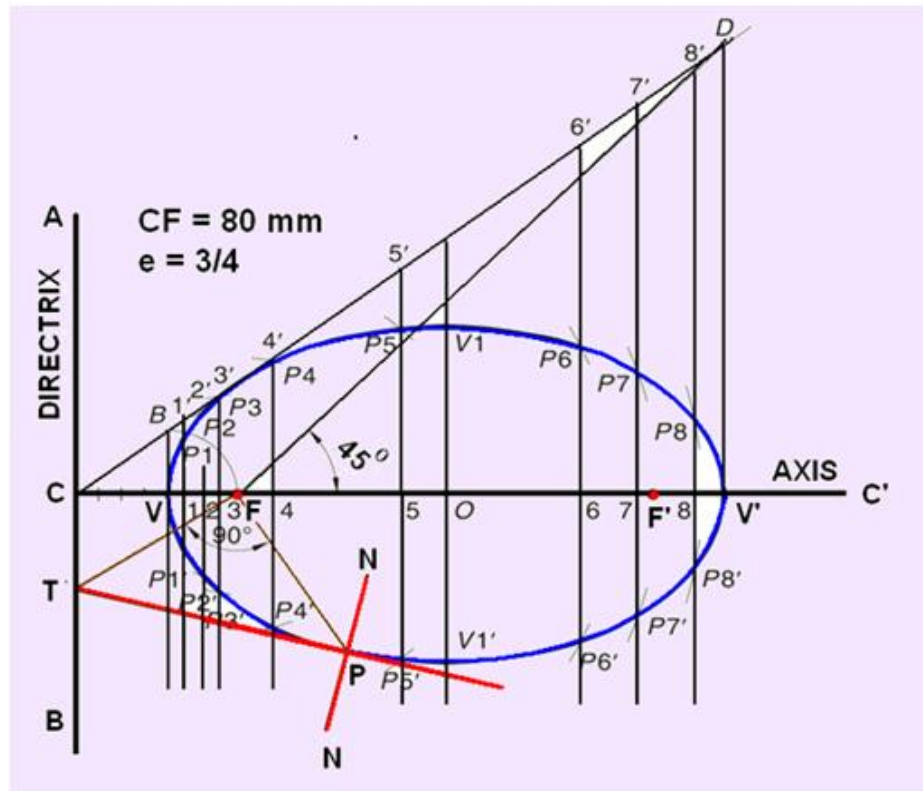


Figure 5. Drawing an ellipse if the distance of focus from the directrix and the eccentricity is given

5. With F as a centre and radius = $1-1'$, cut two arcs on the perpendicular through 1 to locate P_1 and P_1' . Similarly, with F as centre and radii = $2-2'$, $3-3'$, etc., cut arcs on the corresponding perpendiculars to locate P_2 and P_2' , P_3 and P_3' , etc. Also, cut similar arcs on the perpendicular through O to locate V_1 and V_1' .
6. Draw a smooth closed curve passing through V, P_1 , P_2 , P_3 , ..., V_1 , ..., V' , ..., V_1' , ..., P_3' , P_2' , P_1' .
7. Mark F' on CC' such that $V'F' = VF$.

An ellipse is also the set of all points in a plane for which the sum of the distances from the two fixed points (the foci) in the plane is constant. This is clear from figure 6.

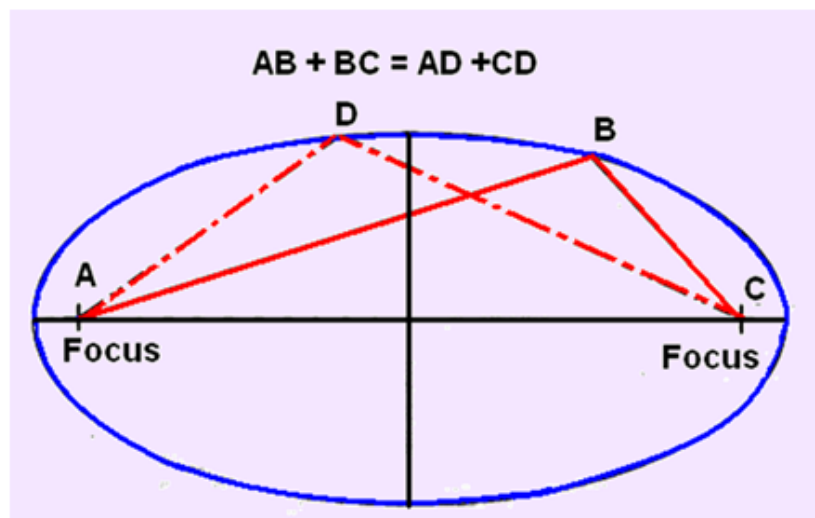


Figure 6. Another definition of ellipse

Arcs of Circle Method:-

The arc of circle method of drawing an ellipse is generally used when (i) the major axis and minor axis are known, and (ii) the major axis and the distance between the foci are known. The method of drawing the ellipse by the arcs of circle method is as follows and is shown in figure 7.

Draw AB & CD perpendicular to each other as the major diameter minor diameter respectively. With centre as C or D, and half the major diameter as radius draw arcs to intersect the major diameter to obtain the foci at X and Y Mark a number of points along line segment XY and number them. Points need not be equidistant.

Set the compass to radius B1 and draw two arcs, with Y as center. Set the compass to radius A1, and draw two arcs with X as center.

Intersection points of the two arcs are points on the ellipse. Repeat this step for all the remaining points. Use the French curve to connect the points, thus drawing the ellipse.

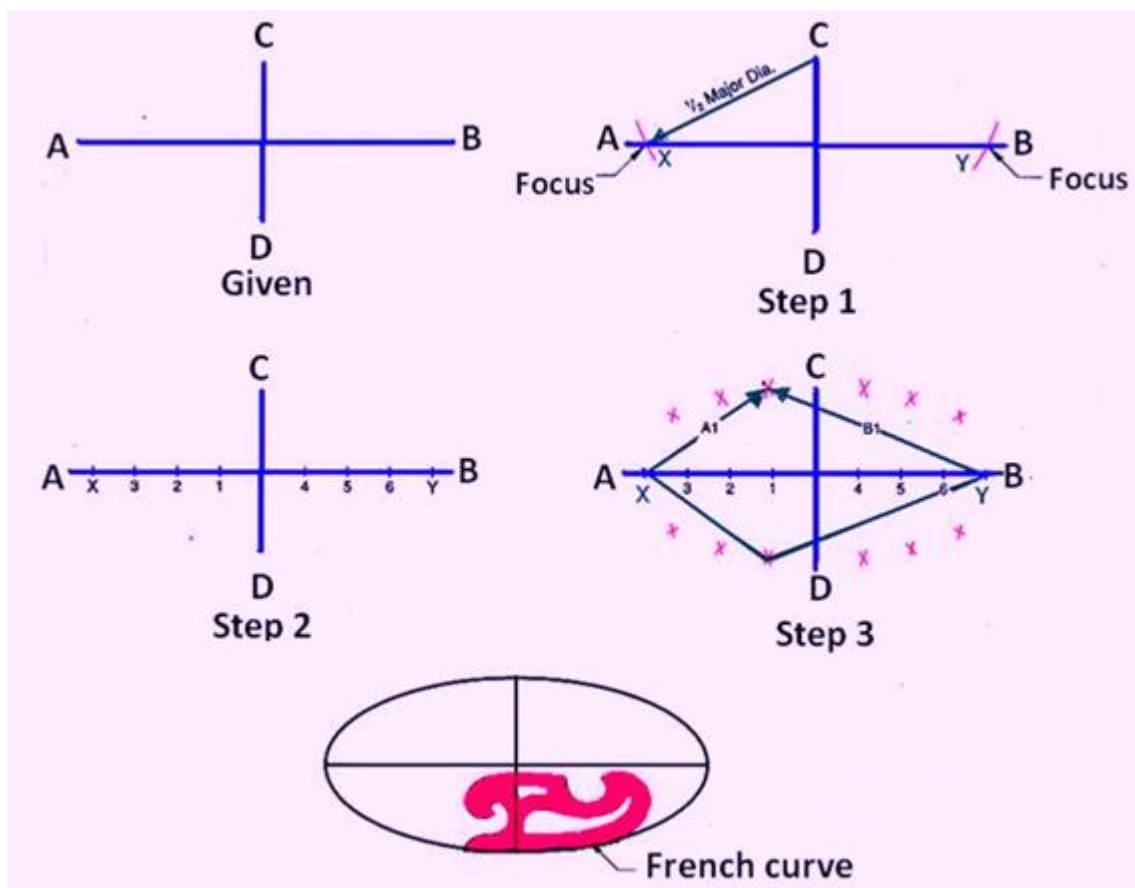


Figure 7. Drawing an ellipse by arcs of circle method.

Constructing an Ellipse (Concentric Circle Method)

- Concentric circle method is used when the major axis and minor axis of the ellipse is given. This method is illustrated in figure 8 and discussed below:
- With center C, draw two concentric circles with diameters equal to major and minor diameters of the ellipse. Draw the major and minor diameters.
- Construct a line AB at any angle through C. Mark points D and E where the line intersects the smaller circle.

- From points A and B, draw lines parallel to the minor diameter. Draw lines parallel to the major diameter through D & E.
- The intersection of the lines from A and D is point F, and from B and E is point G. Points F & G lies on the ellipse.
- Extend lines FD & BG and lines AF and GE to obtain two more points in the other quadrants.
- Repeat steps 2-6 to create more points in each quadrant and then draw a smooth curve through the points.
- With center C, draw two concentric circles with diameters equal to major and minor diameters of the ellipse. Draw the major and minor diameters.

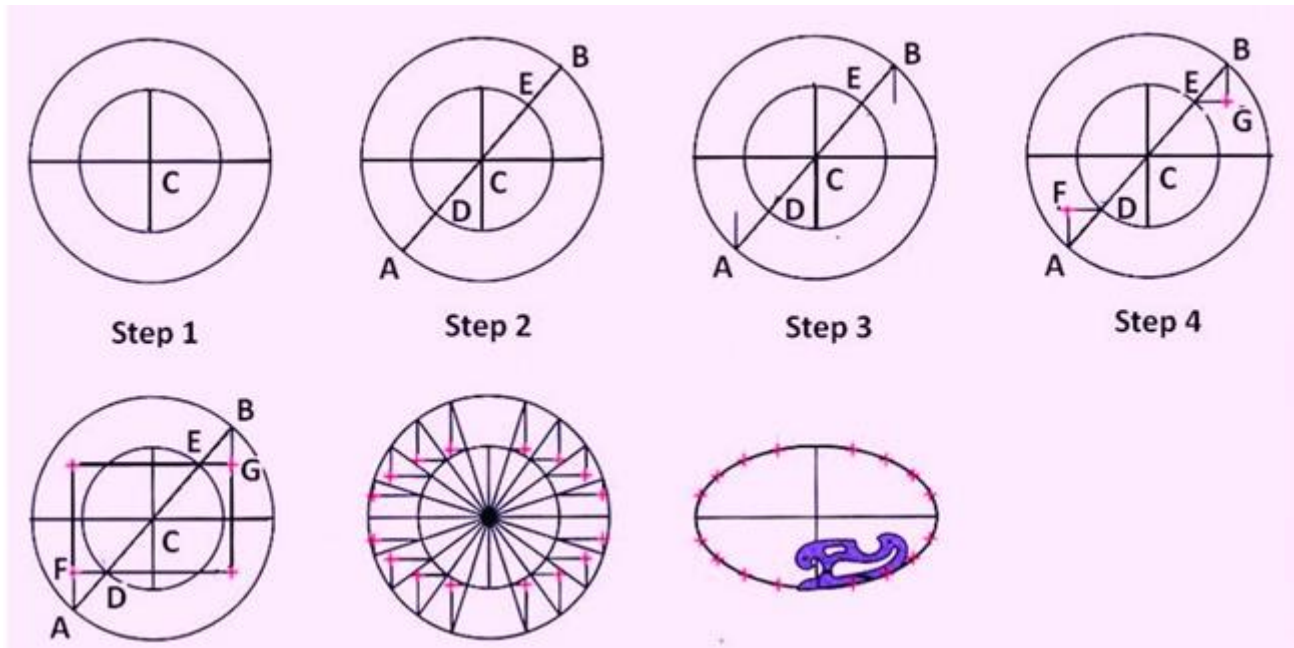


Figure 8. Concentric circle method of drawing ellipse

Drawing Tangent and Normal to any conic

When a tangent at any point on the curve (P) is produced to meet the directrix, the line joining the focus with this meeting point (FT) will be at right angle to the line joining the focus with the point of contact (PF). The normal to the curve at any point is perpendicular to the tangent at that point.

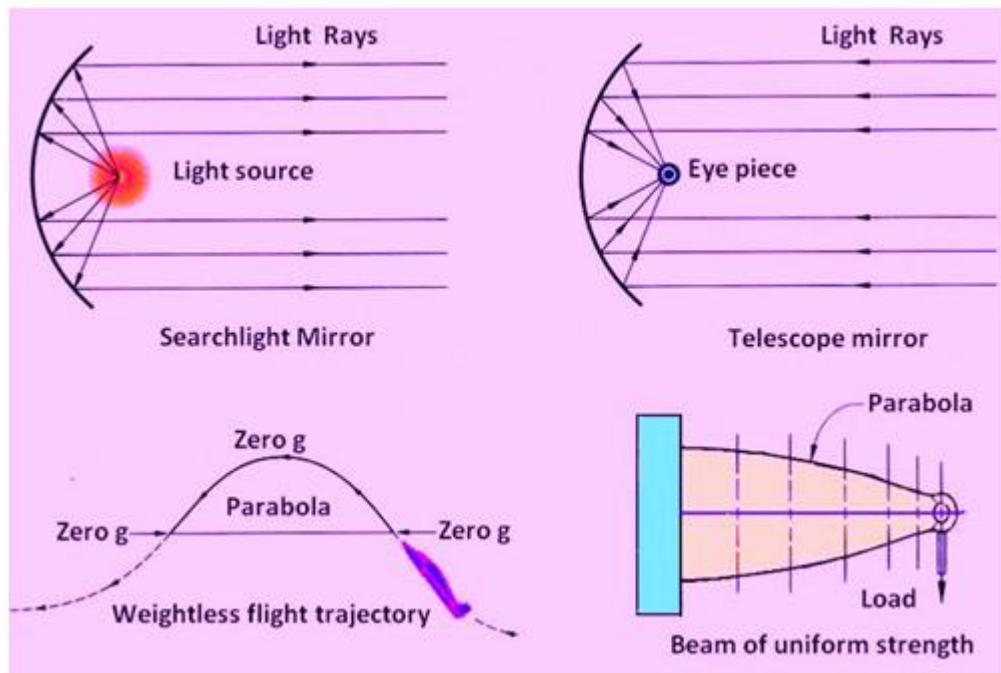


Figure 2. Few applications of parabolic shapes.

Constructing a Parabola (Eccentricity Method)

The method of constructing a parabola by the eccentricity method where the distance of the focus from the directrix is 60 mm is shown in figure 3 and explained below.

1. Draw directrix AB and axis CC' as shown.
2. Mark F on CC' such that CF = 60 mm.
3. Mark V at the midpoint of CF. Therefore, $e = VF / VC = 1$.
4. At V, erect a perpendicular VB = VF. Join CB.
5. Mark a few points, say, 1, 2, 3, ... on VC' and erect perpendiculars through them meeting CB produced at 1', 2', 3', ...
6. With F as a centre and radius = 1-1', cut two arcs on the perpendicular through 1 to locate P1 and P1'. Similarly, with F as a centre and radii = 2-2', 3-3', etc., cut arcs on the corresponding perpendiculars to locate P2 and P2', P3 and P3', etc.
7. Draw a smooth curve passing through V, P1, P2, P3 ... P3', P2', P1'.

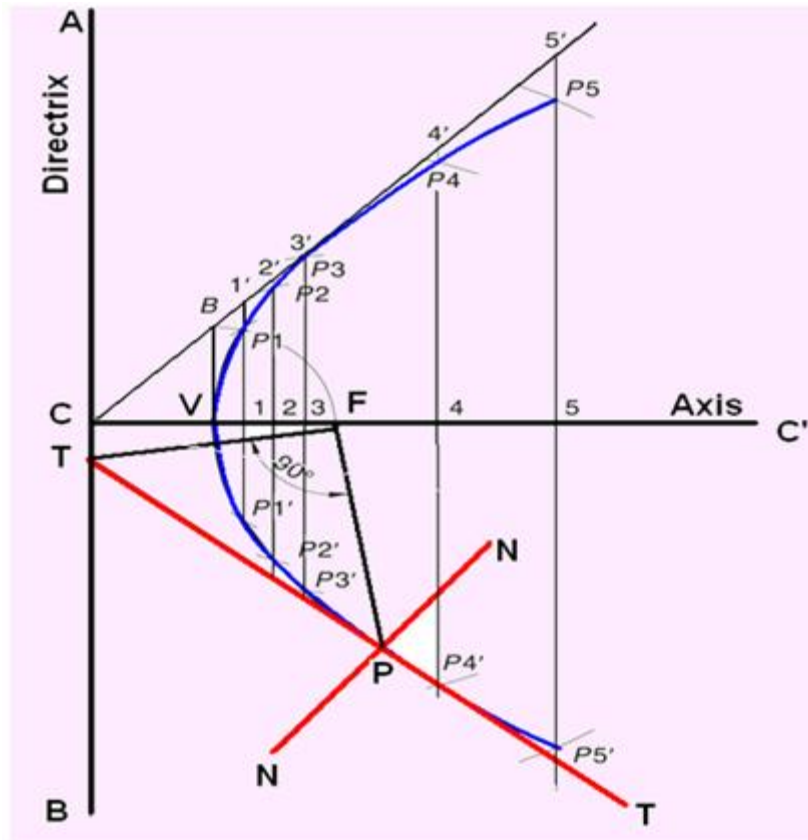


Figure 3. Construction of parabola by eccentricity method.

Constructing a Parabola (Parallelogram Method)

Parabola can also be constructed by parallelogram method. This is illustrated by the example below and shown in fig.4

Example: Draw a parabola of base 100 mm and axis 50 mm if the axis makes 70° to the base.

1. Draw the base RS = 100 mm and through its midpoint K, draw the axis KV = 50 mm, inclined at 70° to RS. Draw a parallelogram RSMN such that SM is parallel and equal to KV.
2. Divide RN and RK into the same number of equal parts, say 5. Number the divisions as 1, 2, 3, 4 and 1', 2', 3', 4', starting from R.
3. Join V-1, V-2, V-3 and V-4. Through 1', 2', 3' and 4', draw lines parallel to KV to meet V-1 at P1, V-2 at P2, V-3 at P3 and V-4 at P4, respectively.
4. Obtain P5, P6, P7 and P8 in the other half of the rectangle in a similar way. Alternatively, these points can be obtained by drawing lines parallel to RS through P1, P2, P3 and P4. For example, draw P1-P8 such that P1-x = x-P8. Join P1, P2, P3 ... P8 to obtain the parabola.

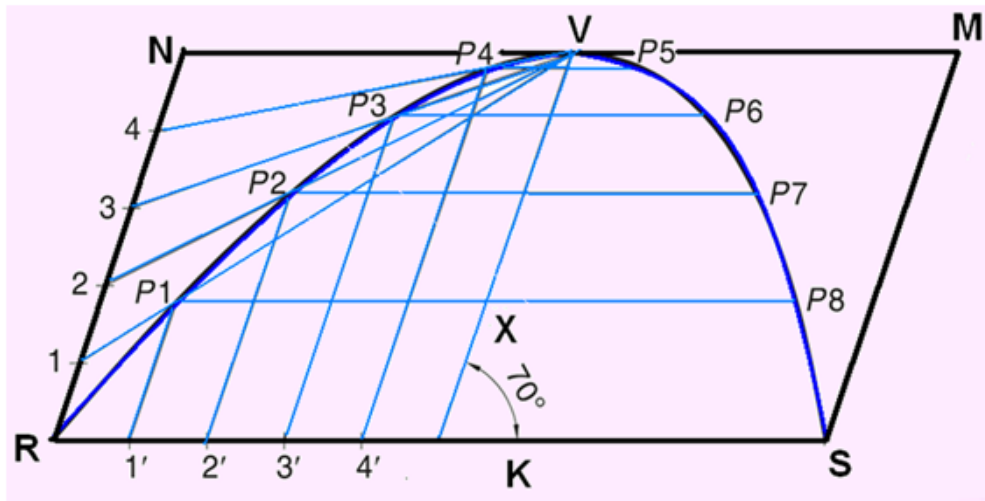


Figure 4. Construction of parabola by parallelogram method.

Tangent Method

This method can be used when the base and the axis, or base and the inclinations of tangents at open ends of the parabola with the base are given. The method is shown in figure 5.

This method can be used when the base and the axis, or base and the inclinations of tangents at open ends of the parabola with the base are given. The method is shown in fig 5.

- Draw the line AB representing the base of the parabola.
- Draw the axis EF representing the height of the parabola.
- Produce EF to O such that $EF = OF$
- Join OA and OB
- Divide OA and OB in to the same number of parts say 8.
- Draw the lines joining 1 to 1', 2 to 2', 3 to 3' etc
- Draw a curve starting from A and tangent to the lines 1-1', 2-2', 3-3', etc, which is the required parabola.

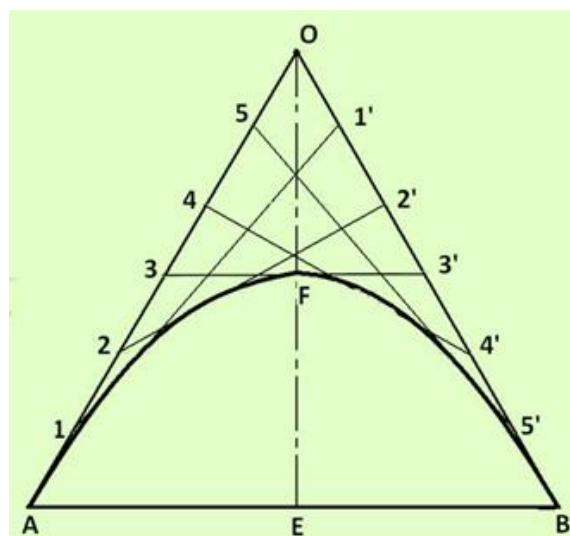


Figure 5. Tangent method of drawing a parabola.

Hyperbola

A Hyperbola is obtained when a section plane, parallel/inclined to the axis cuts the cone on one side of the axis. This is illustrated in figure 1.

A Rectangular Hyperbola is obtained when a section, parallel to the axis cuts the cone on one side of the axis.

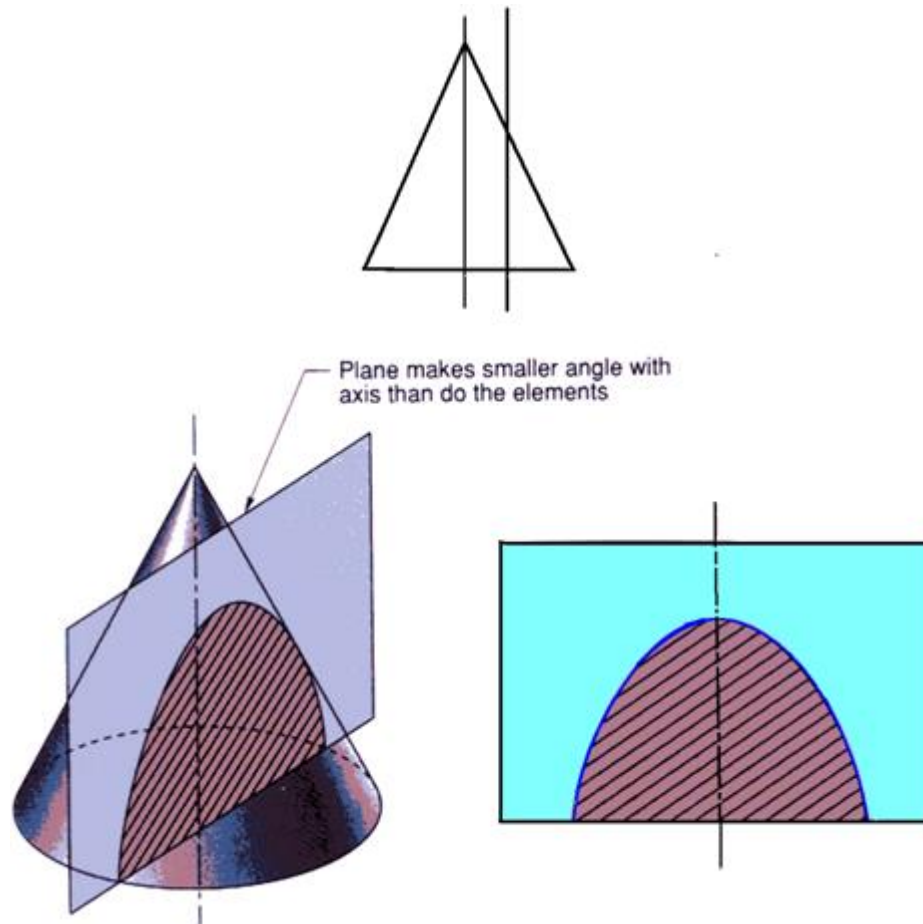


Figure 1. Illustration of a hyperbola.

Hyperbola (Applications)

Hyperbolic shapes find a large number of industrial applications like the shape of cooling towers, mirrors used for long distance telescopes, etc. (as shown in figure 2)

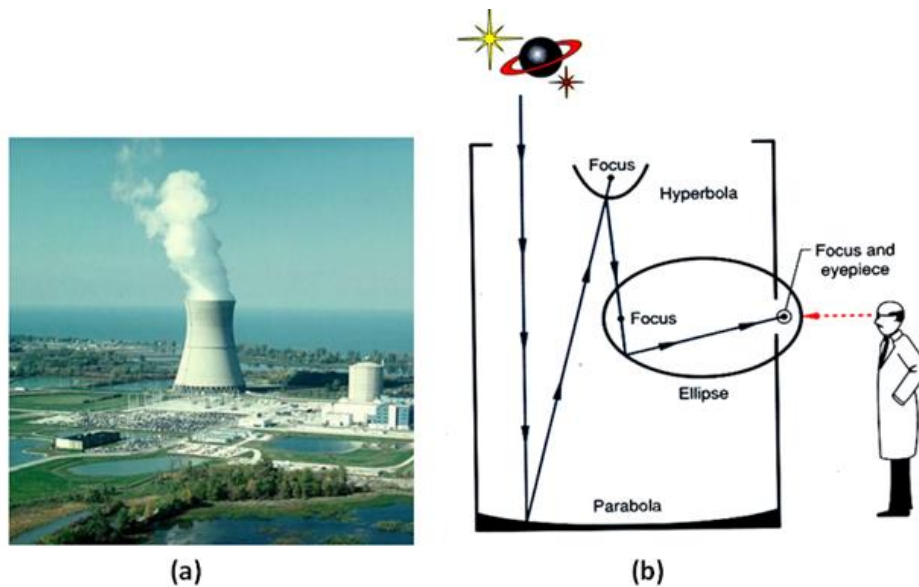
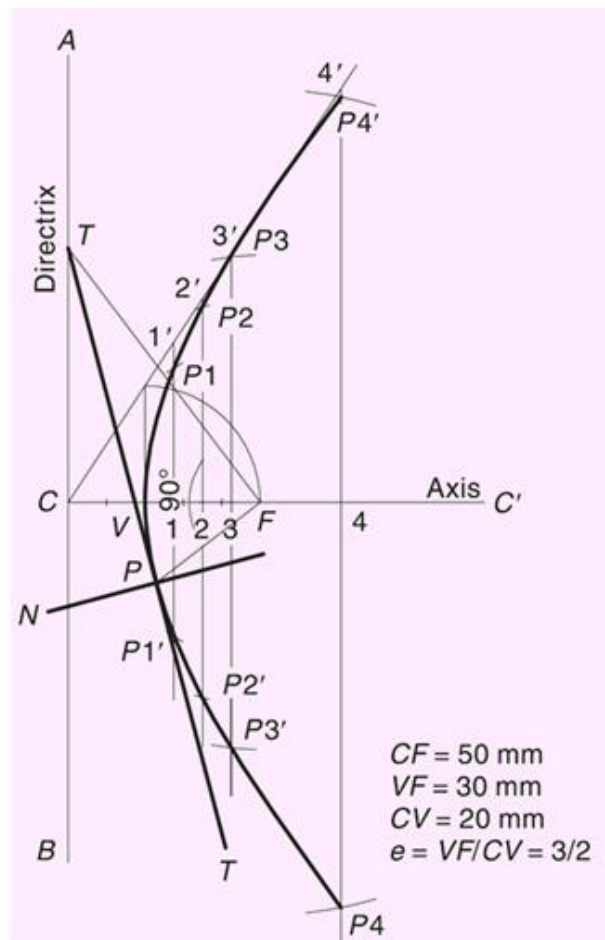


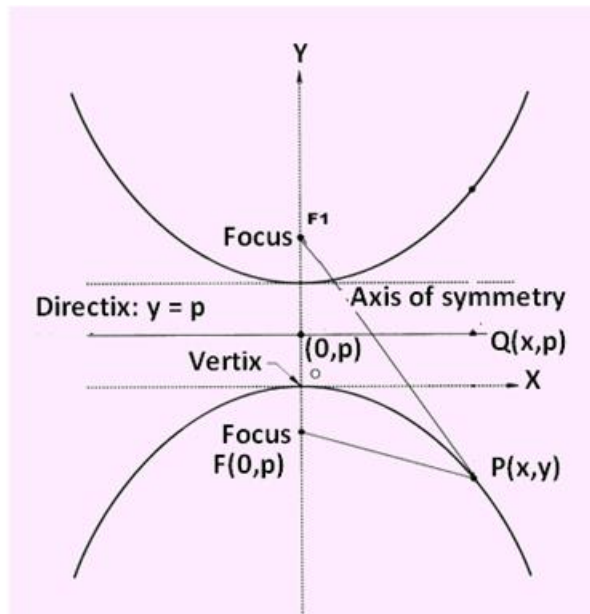
Figure 2. Use of hyperbolic shapes in engineering applications. (*source: internet*)

Constructing a Hyperbola (Eccentricity Method):

Construction of a hyperbola by eccentricity method is illustrated in figure 3, where the eccentricity, $e = 3/2$ and the distance of the focus from the directrix = 50 mm.



A hyperbola is mathematically defined as the set of points in a plane whose distances from two fixed points called foci, in the plane have a constant difference.



Constructing a Hyperbola

Hyperbola can also be construct if the distance between Foci and Distance between vertices are known. This is illustrated in figure 4:

Draw the axis of symmetry and construct a perpendicular through the axis. Locate focal point F equidistant from the perpendicular and on either side of it. Locate points A and B on the axis equidistant from the perpendicular. AB is the distance between vertices.

With F as center and radius R_1 , and draw the arcs. With $R_1 + AB$, radius, and F as center, draw a second set of arcs. The intersection of the two arcs on each side of the perpendicular are points on the hyperbola.

Select a new radius R_2 and repeat step 2. Continue this process until several points on the hyperbola are marked

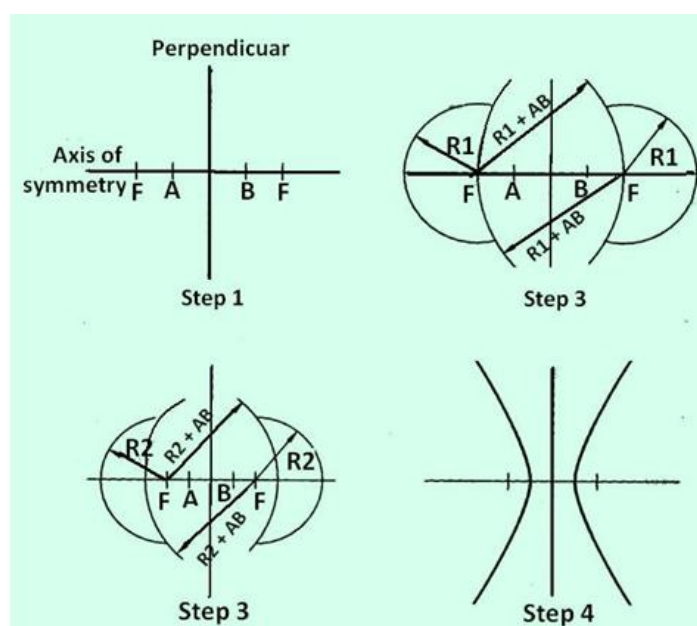


Figure 4. Construction of a hyperbola

Scales

- There is a wide variation in sizes for engineering objects. Some are very large (eg. Aero planes, rockets, etc)Some are very small (wrist watch, MEMs components)
- There is a need to reduce or enlarge while drawing the objects on paper. Some objects can be drawn to their actual size. The proportion by which the drawing of an object is enlarged or reduced is called the scale of the drawing.

Definition:

A scale is defined as the ratio of the linear dimensions of the object as represented in a drawing to the actual dimensions of the same.

- Drawings drawn with the same size as the objects are called full sized drawing.
- It is not convenient, always, to draw drawings of the object to its actual size. e.g. Buildings,
- Heavy machines, Bridges, Watches, Electronic devices etc.

Hence scales are used to prepare drawing at

- Full size
- Reduced size
- Enlarged size

BIS Recommended Scales are shown in table 1.

Table 1. The common scales recommended.

Reducing scales 1:Y (Y>1)	1:2 1:20 1:200 1:2000	1:5 1:50 1:500 1:5000	1:10 1:100 1:1000 1:10000
Enlarging scales X:1 (X>1)	50:1 5:1	20:1 2:1	10:1
Full size scales			1:1

Intermediate scales can be used in exceptional cases where recommended scales cannot be applied for functional reasons.

Types of Scale:-

Engineers Scale: The relation between the dimension on the drawing and the actual dimension of the object is mentioned numerically (like 10 mm = 15 m).

Graphical Scale: Scale is drawn on the drawing itself. This takes care of the shrinkage of the engineer's scale when the drawing becomes old.

Types of Graphical Scale:-

- Plain Scale
- Diagonal Scale
- Vernier Scale
- Comparative scale
- Scale of chords

Representative fraction (R.F.):

$$R.F. = \frac{\text{Length of an object on the drawing}}{\text{Actual Length of the object}}$$

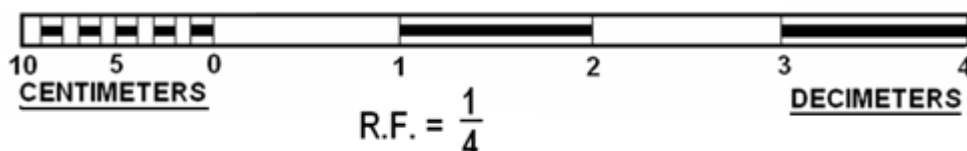
When a 1 cm long line in a drawing represents 1 meter length of the object

$$R.F. = \frac{1\text{cm}}{1\text{m}} = \frac{1\text{cm}}{1 \times 100\text{cm}} = \frac{1}{100}$$

Length of scale = RF x Maximum distance to be represented

Plain scale:-

- A plain scale is used to indicate the distance in a unit and its next subdivision.
- A plain scale consists of a line divided into suitable number of equal units. The first unit is subdivided into smaller parts.
- The zero should be placed at the end of the 1st main unit.
- From the zero mark, the units should be numbered to the right and the sub-divisions to the left.
- The units and the subdivisions should be labeled clearly.
- The R.F. should be mentioned below the scale.

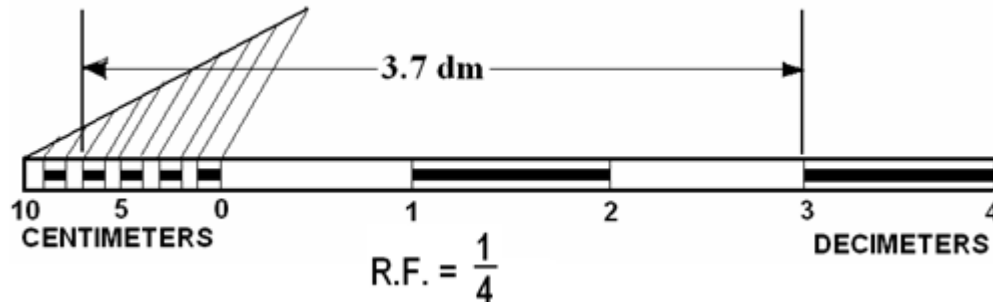


Construct a plain scale of RF = 1:4, to show centimeters and long enough to measure up to 5 decimeters.

R.F. = $\frac{1}{4}$

- Length of the scale = R.F. $\times$ max. Length = $\frac{1}{4} \times 5 \text{ dm} = 12.5 \text{ cm}$.
- Draw a line 12.5 cm long and divide it into 5 equal divisions, each representing 1 dm.
- Mark 0 at the end of the first division and 1, 2, 3 and 4 at the end of each subsequent division to its right.

- Divide the first division into 10 equal sub-divisions, each representing 1 cm.
- Mark cm to the left of 0 as shown.
- Draw the scale as a rectangle of small width (about 3 mm) instead of only a line.
- Draw the division lines showing decimeters throughout the width of the scale.
- Draw thick and dark horizontal lines in the middle of all alternate divisions and sub-divisions.
- Below the scale, print DECIMETERS on the right hand side, CENTIMETERS on the left hand side, and R.F. in the middle.

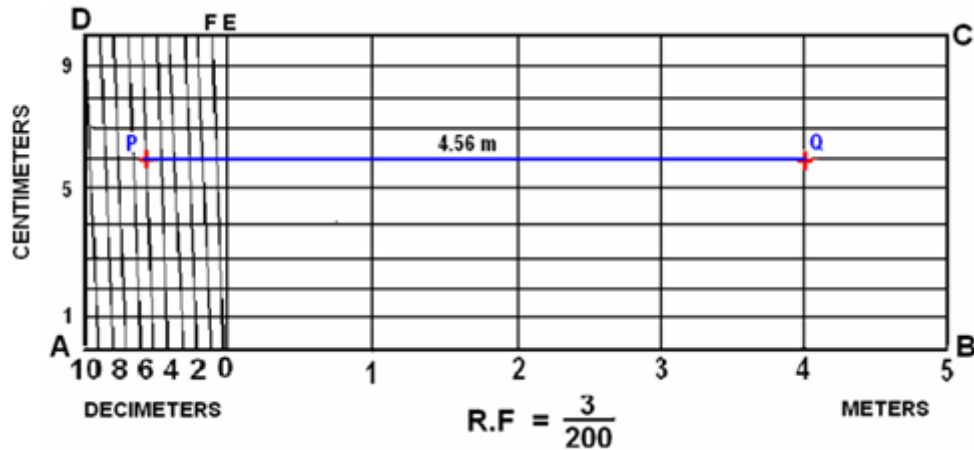


Diagonal Scale:-

- Through Diagonal scale, measurements can be up to second decimal places (e.g. 4.35).
- Are used to measure distances in a unit and its immediate two subdivisions; e.g. dm, cm & mm, or yard, foot & inch.
- Diagonal scale can measure more accurately than the plain scale.
- Diagonal scale.....Concept
- At end B of line AB, draw a perpendicular.
- Step-off ten equal divisions of any length along the perpendicular starting from B and ending at C.
- Number the division points 9, 8, 7,.....1.
- Join A with C.
- Through the points 1, 2, 3, etc., draw lines parallel to AB and cutting AC at 1', 2', 3', etc.
- Since the triangles are similar; $1'1 = 0.1 AB$, $2'2 = 0.2AB$, $9'9 = 0.9AB$.
- Gives divisions of a given short line AB in multiples of $\frac{1}{10}$ its length, e.g. $0.1AB$, $0.2AB$, $0.3AB$, etc.

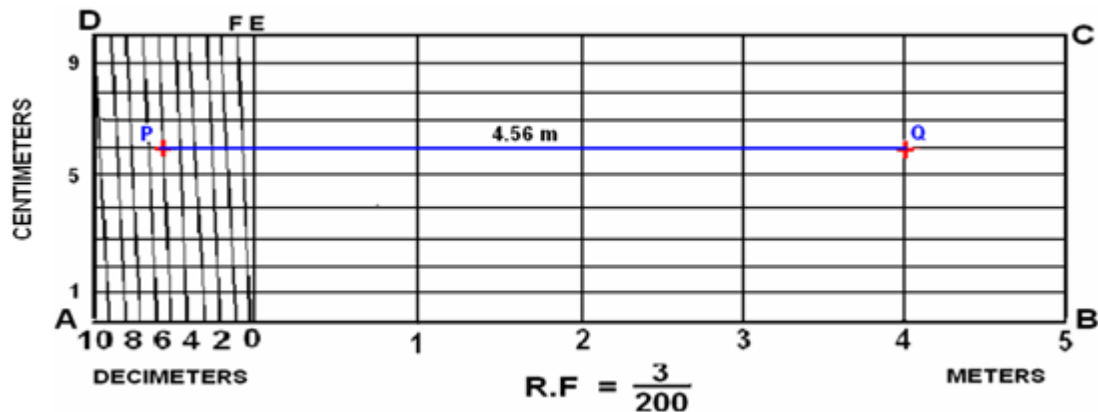


Construct a Diagonal scale of RF = 3:200 showing meters, decimeters and centimeters. The scale should measure up to 6 meters. Show a distance of 4.56 meters



- Length of the scale = $(3/200) \times 6 \text{ m} = 9 \text{ cm}$
- Draw a line AB = 9 cm . Divide it in to 6 equal parts.
- Divide the first part A0 into 10 equal divisions.
- At A draw a perpendicular and step-off along it 10 equal divisions, ending at D.

Diagonal Scale



- Complete the rectangle ABCD.
- Draw perpendiculars at meter-divisions i.e. 1, 2, 3, and 4.
- Draw horizontal lines through the division points on AD. Join D with the end of the first division along A0 (i.e. 9).
- Through the remaining points i.e. 8, 7, 6, ... draw lines // to D9.
- PQ = 4.56 meters
- Theory of Projections